

Received 21 October 2025, accepted 2 November 2025, date of publication 10 November 2025,
date of current version 14 November 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3631323

RESEARCH ARTICLE

An Affordance Perspective on Generative AI in Enterprise Architecture

LEONARDO BANH^{ID}, TUAN KHANG TRAN, AND GERO STROBEL^{ID}

University of Duisburg–Essen, 45141 Essen, Germany

Corresponding author: Leonardo Banh (leonardo.banh@uni-due.de)

This work was supported by the Open Access Publication Fund of the University of Duisburg–Essen.

ABSTRACT Generative artificial intelligence (GenAI) has emerged as a transformative technology capable of creating novel content across multiple formats. Its rapid advancement fuels innovations and implications for various fields, including the unexplored domain of enterprise architecture. This study investigates the potential applications of generative AI in enterprise architecture through semi-structured expert interviews, analyzed using the theoretical lens of affordance theory. The research identifies four key affordances that generative AI offers enterprise architects: information research and synthesis, text generation and refinement, insight generation and decision support, and architectural content creation. The findings reveal that generative AI functions as a collaborative partner for enterprise architects, providing information and inspiration that, when combined with professional knowledge and expertise, enables the creation of high-quality and innovative enterprise architecture content. This technology shows promise in enhancing productivity, efficiency, and effectiveness in enterprise architecture practices.

INDEX TERMS Generative artificial intelligence, enterprise architecture management, human-GenAI collaboration, affordance theory, qualitative interview study.

I. INTRODUCTION

In today's world, artificial intelligence (AI) is ubiquitous and permeates a wide range of areas, from personalized experiences provided by social media platforms or online retailers to autonomous vehicles and smart home assistants [1], [2], [3]. AI has become such an integral part of daily life that it is difficult to imagine everyday processes without the support of this technology [4], [5]. Even companies are now aware of the economic importance of AI, as this technology is increasingly becoming a competitive factor [6], [7]. Over 70 % of technology leaders agree that AI has already moved beyond the experimental phase and is now generating real business value [8]. The most recent AI milestone is the emergence of generative AI (GenAI) as a novel and paradigm-changing type of AI [9], [10]. A leading example is ChatGPT, a multi-purpose GenAI system that has revolutionized human-machine interaction and is capable of providing assistance with a wide range of everyday tasks, including creating text, translating it into different languages, solving

coding problems, or extracting information on various topics [9], [11], [12]. Just two months after its launch, ChatGPT already reached 100 million monthly active users, making it the fastest growing consumer application in history [13]. Ongoing research and development regarding GenAI demonstrate its transformative potential for a wide range of sectors, emphasizing the need to streamline processes and increase competitiveness in the economic landscape [14], [15], [16], [17]. While text-generating chatbots like ChatGPT have received considerable attention, GenAI can enable capabilities for a wide range of content including images, videos, audio, and computer code [9]. These GenAI models are designed to learn patterns, structures, and distributions for the occurrence of certain features in the real-world data provided, and based on this generate new synthetic output data with similar characteristics [18].

The market size of generative AI is expected to register a compound annual growth rate of 25% from 2023 to 2030, leading to a market size of \$207 billion by 2030 [19]. This growth indicates that companies are expected to continuously invest in GenAI to benefit from its multiple applications. In addition, the technology has the potential to become a

The associate editor coordinating the review of this manuscript and approving it for publication was Ugur Guvenc^{ID}.

crucial and sustainable technology trend that could further expand in the future and fundamentally change various industries [20]. For instance, GenAI in software engineering is able to support code generation and completion, as well as helping to improve the understanding of existing code to gain a productivity increase of up to 50% [21], [22], [23]. In the medical field, GenAI can help doctors to diagnose diseases based on medical records, images and lab results, additionally suggesting treatment options or plans, improving the accuracy and efficiency of medical treatment [24], [25].

Although the applications of this technology are constantly expanding, the full potential has not yet been exhausted, and new possibilities and fields of application are still being intensively investigated. One potential application area of GenAI is in the context of enterprise architecture (EA) where the field could benefit from the supporting capabilities of GenAI to assist enterprise architects in their activities [26]. EA comprises a holistic view on business processes, applications, and the underlying IT infrastructure, with optional inclusion of business units, data architecture, software architecture, and IT products [27]. The goal of EA is to guide the design and development of an organization's operational platform and encompass IT systems and digitized business processes that support or automate an organization's core capabilities [28].

To remain competitive, expand growth or increase resilience to competitors, companies need to continuously improve their products, services, business operations, and technologies [29]. However, these frequent changes have a significant impact on the underlying EA, which can be affected by changing operational requirements or external factors. These adaptations commonly involve changing organizational responsibilities, refining business processes and fixing core information system errors [27]. Nevertheless, the complexity and scale of enterprise modernization often presents EA practitioners with significant challenges. Due to their supporting capabilities, such as reducing time spent on low-value tasks, GenAI has the potential to enhance productivity, efficiency, and effectiveness for enterprise architects utilizing this technology in their activities [7], [30], [31]. This objective solution presents a promising future for increasing business performance and streamlining processes in EA.

Despite its promises, research on GenAI in the context of EA is still in its infancy. In the general field of AI, Rittelmeyer and Sandkuhl [32] have reviewed the impact of implementing AI applications in EA, concluding that AI could impact both EA activities (e.g., substitution of workflows and functions) and the modelling of EA itself (e.g., change of designing software architectures). To facilitate successful implementation, widespread adoption, and improved communication between the business and technology sides, it is important to integrate a dedicated AI layer into the architecture [32], [33]. However, research on the impact of AI on EA has been kept at a high level, whereas concrete details on how exactly EA activities are influenced by AI or how different parts of the enterprise architecture could be specifically influenced are overlooked. Specific examples or case studies could provide

a better understanding of the actual impact of AI on EA, thus offering research potential to explore this relevant topic [34].

To initiate research and close the current lack of knowledge, it appears appropriate to identify possible applications of GenAI in the context of EA. By employing affordance theory as theoretical lens, it is possible to investigate the perceived possibilities of use for the actions that enterprise architects associate with adopting and leveraging GenAI [35]. Furthermore, there is an imperative need to improve comprehension regarding how GenAI can be assimilated into EA operations to accomplish precise objectives [34], [36]. This awareness is essential in guiding organizations to determine which areas of EA they can target for the use of GenAI to benefit from technological advances. Consequently, this opens the potential for data-driven theory building to better explain and understand the impact of EA through generative AI. Against this background, we address the following research questions:

RQ1: What affordances does generative AI provide to enterprise architects?

RQ2: How can enterprise architects leverage generative AI to support their activities?

To answer these questions, we chose a qualitative research approach and conducted an interview study with experts in EA to develop a theoretical framework on the application of GenAI. We developed a semi-structured interview guide that focused on the elicitation of the experts' thoughts and views on GenAI for their EA-related tasks [37]. Applying affordance actualization theory [35], [38] as theoretical lens during the data analysis and coding allowed us to focus on how GenAI facilitates EAs' actions on their job, synthesizing their goals and values, and sketching out how the technology can be integrated into organizational and individual routines and practices. Hence, this study offers a grounded framework explaining how GenAI supports EA activities and which potentials and challenges affect individual and organizational actions from a socio-technical perspective. The insights provide a useful foundation for future researchers who want to investigate the adoption of GenAI as well as for EA practitioners who seek to apply GenAI for improving their EA management activities.

This paper is structured as follows: First, theoretical foundations regarding GenAI in EA as well as affordance theory are introduced in Chapter 2. Subsequently, Chapter 3 describes the research methodology, and Chapter 4 presents the results of our study. Finally, the results are discussed in Chapter 5, before the paper culminates with a conclusion in Chapter 6.

II. THEORETICAL BACKGROUND

A. APPLYING GENERATIVE AI TO ENTERPRISE ARCHITECTURE MANAGEMENT

Recent advancements in AI have catalyzed the emergence of GenAI, a class of AI systems capable of producing realistic, coherent, and context-aware outputs across a wide range of

modalities, including text, diagrams, and structured data [9], [39]. Unlike traditional discriminative models, which learn mappings from inputs to labels, deep generative models, such as variational autoencoders, generative adversarial networks, and large transformer-based language models, learn high-dimensional probability distributions, enabling them to generate novel, domain-consistent outputs [40], [41]. As a result, GenAI supports use cases that go beyond traditional classification or prediction tasks, offering synthetic generation of artifacts such as architecture diagrams, textual requirements, or compliance reports, all of which have historically demanded extensive manual effort in the field of EA [42], [43], [44].

A well-implemented and well-documented EA can benefit an organization in various ways. These benefits include strategic enterprise transformation, cost reduction through the elimination of redundant technologies, a holistic business view, better business-IT alignment, optimized change management, efficient risk management, improved business function interoperability and integration, and easier management of IT resource complexity [28], [45].

The application of GenAI in EA introduces a novel interaction paradigm that enables practitioners to engage with architecture models and information systems through natural language interfaces and co-creative processes [26], [46], [47]. For example, large language models (LLMs) trained on EA repositories can map strategic goals to application-and-technology components, automate traceability links across business, data, application, and technology layers, and support the generation of enterprise models [36], [48], [49]. Such capabilities directly address two critical EA pain points: (i) the labor-intensive creation and maintenance of multi-view models, and (ii) the cognitive burden of aligning heterogeneous stakeholder perspectives [34], [44], [50], [51].

Beyond artifact generation, GenAI facilitates conversational modeling assistants that engage architects in a co-creative loop: translating natural-language queries into model constructs, suggesting alternative designs, or simulating “what-if” scenarios under evolving business contexts [32], [48], [52]. Preliminary prototypes demonstrate that this human-in-the-loop paradigm can improve both modeling consistency and stakeholder engagement, while freeing architects to focus on high-level strategy [53], [54]. This cognitive augmentation mechanism allows architects to shift focus from manual modeling to higher-level analysis and strategic decision-making [55]. As a result, enterprise architects can gain increased efficiency and consistency, particularly in complex or federated enterprise environments where architectural knowledge is distributed and heterogeneous [56], [57].

Despite these promising capabilities, the integration of GenAI in EA remains nascent and introduces its own challenges. For instance a primary concern is the explainability and trustworthiness of generated outputs, especially given the formal and regulatory nature of enterprise

systems [58]. Errors in generated models, misalignment with established EA frameworks, or hallucinated architectural elements may lead to flawed decision-making or non-compliant designs [46]. Moreover, GenAI’s reliance on high-quality data and effective prompting limits its effectiveness in organizations with fragmented or undocumented architecture data [59], [60].

In addition, educational and organizational readiness are increasingly recognized as critical enablers for GenAI adoption in EA [34], [44], [61]. Similar to the domain of software engineering, a lack of critical evaluation skills or overreliance on GenAI outputs may result in superficial models or suboptimal decisions that hinders effective adoption in work processes [22], [62]. Training EA practitioners to effectively collaborate with AI systems, understand their limitations, and apply architectural judgment remains essential for leveraging the full potential of GenAI [9], [63].

Consequently, GenAI represents a transformative innovation in enterprise architecture management by introducing new capabilities for model generation, knowledge synthesis, and decision support [64]. By augmenting the architectural process with generative capabilities and natural language interaction, GenAI can reduce documentation burdens, support strategic alignment, and enable more adaptive and responsive enterprise design. However, realizing this potential requires careful attention to sociotechnical factors and to foster its integration into established EA frameworks and organizational contexts.

B. AFFORDANCE ACTUALIZATION THEORY

Affordance theory has become a well-established and widely used vehicle for conceptualizing the affordances that a technical artifact provides to an actor [35], [65]. It provides researchers with a variety of research opportunities, enabling intensive engagement with technology and inclusion of the social context in investigations [66]. According to Volkoff and Strong [35], affordances are defined “as the potential for behaviors associated with the achievement of an immediate concrete outcome that result from the relationship between an object (e.g. an IT artefact) and a goal-oriented actor” [35]. By applying affordance theory, it is therefore possible to investigate the perceived possibilities for actions that a goal-directed user associates with technology. However, affordances arising from the relationship between an artifact and a goal-oriented actor cannot only be considered and analyzed in the context of individual persons but also an entire organization or a specific group within that organization [35].

When identifying affordances, it is essential to recognize that an IT artifact presents users with various affordances, which are bundles of possibilities. These affordances can also be linked and interdependent in different ways [66]. For example, the implementation of an electronic health record system enables analyzing patient data, although this requires that the patient data has been previously recorded and stored in the system [67].

In this study, we leverage affordance theory to identify potential actions for GenAI in the context of EA. Despite being considered a valuable tool, affordance theory has focused more on how an actor perceives an object in its environment and less on what happens when the actor interacts with the object after perception [66]. Therefore, affordance actualization theory extends the original affordance theory by distinguishing between the use of technology, its affordances, and the resulting effects to create a comprehensive understanding of how specific technology features provide affordances for users and what the resulting outcomes are [66], [68]. Hence, it is highlighted that the mere existence of an affordance does not automatically lead to its realization [69]. In order to achieve the desired benefits, users must actively realize the possibilities of a technology [70].

Within the context of EA, an affordance can be seen as an actionable possibility that emerges from the interplay between GenAI's technological capabilities (e.g., LLM-driven synthesis, classification, and dialogue) and the goals, skills, and practices of EA professionals, mediated by organizational context and constraints. Specifically, GenAI affordances in EA refer to the novel ways in which GenAI systems can support, augment, or automate EA design, analysis, and governance tasks when these systems are recognized and actualized by users.

We focus on the individual application of affordances, exploring the possibilities that GenAI technologies offer to enterprise architects and how these possibilities are put into practice to support them in their activities. It is important to mention that the implementation of an affordance is not a linear path that leads to a desired goal if implemented successfully. It can also be an iterative process, leading to the discovery of new affordances as potentially unpredictable outcomes are achieved, which in turn increases awareness and understanding of the affordances of the technology itself [69]. The existence of the affordances is not influenced by the actor's consciousness. However, the extent to which the actor is aware of them can affect the actualization of the affordance [66]. This can mean that affordances are not realized because the user is not aware of them. Alternatively, it can indicate that an actor realizes an affordance without being aware of it.

This lack of awareness may result in ineffective use of the IT artifact, even if the potential for action exists [66]. Furthermore, the implementation of an affordance can enhance the user's knowledge, enabling them to use the technology more effectively and proficiently by adapting their actions accordingly [67], [71].

In addition to the differentiation between the affordances and their realization, affordance actualization theory also considers the organizational perspective [72]. The organizational context holds particular importance in this context, as it can facilitate or even limit the process of affordance realization [67], [71]. Therefore, companies must also meet certain requirements to ensure the effective realization of affordances.

In this work, we identify and examine the affordance actualization process that GenAI brings to EA practice, such as the ability to rapidly synthesize candidate architectures, support the co-creation of EA models through conversational agents, or uncover architectural patterns across heterogeneous artifacts. Each instance illustrates how material GenAI capabilities, when actualized by EA practitioners within their workflow and governance structures, enable new forms of design, analysis, and coordination that were previously impractical or impossible to assist through technology.

III. RESEARCH METHODOLOGY

Qualitative semi-structured expert interviews were utilized as the primary source for **data collection**. This approach is appropriate for gathering detailed, empirical data, particularly when investigating episodic and infrequent phenomena [70], [73]. Expert interviews enable a reciprocal exchange between the interviewer and respondent, enabling additional questions to be asked spontaneously in response to the respondent's answers and providing room for the respondent's unique oral contributions [74]. The semi-structured interview questions were designed using an interview guide to ensure comprehensive coverage of the subject area [75]. The questions for the semi-structured interviews were developed based on the theory of affordance actualization and were designed to capture qualitative information on the key elements of the affordances themselves, their realization, and their outcomes. This approach support the process of unfolding how GenAI can support enterprise architects, since affordances and their realization are not directly observable in themselves, but instead only the resulting effects can be perceived [35], [38]. Exemplary questions employed included “*What are the motivations and reasons for using generative AI in your work activities?*” or “*What EA tasks can be accomplished today with the assistance of generative AI that were previously difficult or impossible?*” The interview guide was refined and specified in several iterative rounds to ensure that data on the core aspects of the affordance realization theory could be collected, i.e., how enterprise architects see, use and envision GenAI for their work. A pre-test of the preliminary semi-structured interview was conducted to identify further difficulties in understanding the questions and determine the necessary interview time [74].

The recruitment of experts was initiated by identifying potential candidates within a single case multi-national company of consulting enterprise architecture experts [76], [77]. Drawing from the concepts of a single case study, investigating a particular case company allows us to attain in-depth, exploratory insights within a global organization and how the disruptive innovation unfolds [76]. By investigating a specific organization, we are able to reveal contextualized insights that would be difficult to spot across broader, multiple sites. Our selected case organization is a global leader in consulting and enterprise architecture services. Its prominent role in shaping enterprise architecture practices made it a theoretically and practically relevant site for examining adoption

mechanisms and boundary conditions in detail. This focused approach enabled us to gain analytic depth regarding the practical realization of GenAI affordances within established EA processes.

The selected individuals were required to have a minimum of five years of experience in the context of EA and to have already utilized GenAI in their professional activities. Furthermore, efforts were made to ensure that these individuals represented a diverse range of EA domains, including strategy, integration, and tooling, for a comprehensive representation of the phenomenon. A total of ten experts were recruited within the case company to participate in this study. Between February and March 2024, these experts were interviewed for between 48 and 72 minutes, with an average of about 52 minutes. Table 1 provides an overview of the experts interviewed.

TABLE 1. Overview of the participating EA respondents.

ID	EA Position	Industry Focus	Duration
01	Strategy Principal Director	Generalist	1 h 12 min
02	EA Consultant	Generalist	52 min
03	Strategy Principal Director	Generalist	45 min
04	Senior Manager Technology	Generalist	41 min
05	Consulting Senior Principal	Healthcare	56 min
06	Senior Manager Technology	TelCo Industrial	51 min
07	Consulting Principal Director	Equipment and Resources	48 min
08	Strategy Leadership	Generalist	52 min
09	Manager	Generalist	52 min
10	Senior Manager	Travel	57 min

At the time of data collection, participants described GenAI as being in an exploratory phase within their organization, with ongoing efforts to build competencies and develop internal products for EA-related use; however, formal anchoring in operational frameworks had not yet occurred, although several respondents noted that management had expressed a desire to establish such integration. The interviews were all conducted virtually, recorded and then transcribed verbatim to enable a detailed data analysis. The data from the ten recorded interviews resulted in approximately 106 single-spaced pages of transcribed conversations.

For **data analysis**, grounded theory methodology (GTM) was used [78], [79], employing a three-stage coding process of open, axial, and selective coding. The method offers a systematic approach to deriving theories from empirical data [80]. It involves a series of procedures in which data are systematically broken down, conceptualized, and reassembled in new ways [81]. This process enables the recognition of deeper patterns and relationships within the data, leading to the development of a new theory closely linked to the empirical material. To simplify the coding and analysis process, the

qualitative data analysis tool MAXQDA was used. The initial stage of the coding process involved open coding, whereby the raw data from the expert interviews were analyzed. This involved linking text passages to concepts. For instance, the statement *“It is helpful to consult GPT and obtain feedback, which saves time. You can do it independently, without having to approach others or bother them. And often, AI is more knowledgeable than a random colleague”* by respondent 02 was coded into the first-order code *“Improving the quality of work by getting initial feedback”*. Following Gioia et al. [79], care was taken in this first-order analysis to maintain the terminology of the data sources. The result was an extensive list of more than 220 descriptive first-order codes (see Appendix A for an excerpt of our data structure [79]). Adhering to the concept of GTM’s constant comparison, open coding was performed iteratively during the data collection until no new insights emerged. Hence, theoretical saturation ended the recruitment of additional interview participants. To ensure internal reliability and achieve consistent coding themes, open coding was performed independently within the author team, and any disagreement was resolved collectively through discussions during the process.

In the second stage of the coding process, axial coding was initiated after reaching the data saturation point when no new concepts could be identified. In line with GTM proposed by Gioia et al. [79], second-order analysis identified similarities and differences between the various codes. First-order codes that related to the same phenomenon and helped to explain the phenomenon were grouped into superordinate themes. Then, labels were assigned to the overarching themes, taking care to preserve the terms used by the experts and ensure that the labels reflected the content of the categories they contained. A total of 42 themes were derived from the existing codes. In addition to the grouping of codes with similar content, relationships between codes were also identified during the axial coding phase. Continuing with the previous example, the second-order theme *“Quality improvement of work results”* was identified that also captures other first-order codes such as *“Quality assurance through critical reflection on content and perspectives”*.

In the final stage of the three-step coding process, selective coding was applied by synthesizing extant secondary themes into a set of aggregated dimensions. These dimensions were deductively derived from essential components of the affordance actualization theory [66] and included the elements such as goal-oriented actor, affordances, outcomes, and organizational context. Thus, the coding example of quality improvement was assigned to the aggregate dimension *“Goal-oriented actor”* as it described the objectives of enterprise architects when choosing to use GenAI. Affordance theory therefore guided the interpretation and theorization process during data analysis as a sensitizing and explanatory lens to structure how and why GenAI is used in EA. Also, the dimensions of *“qualification of enterprise architects”* and the *“enrichment of generative AI outcomes”* emerged

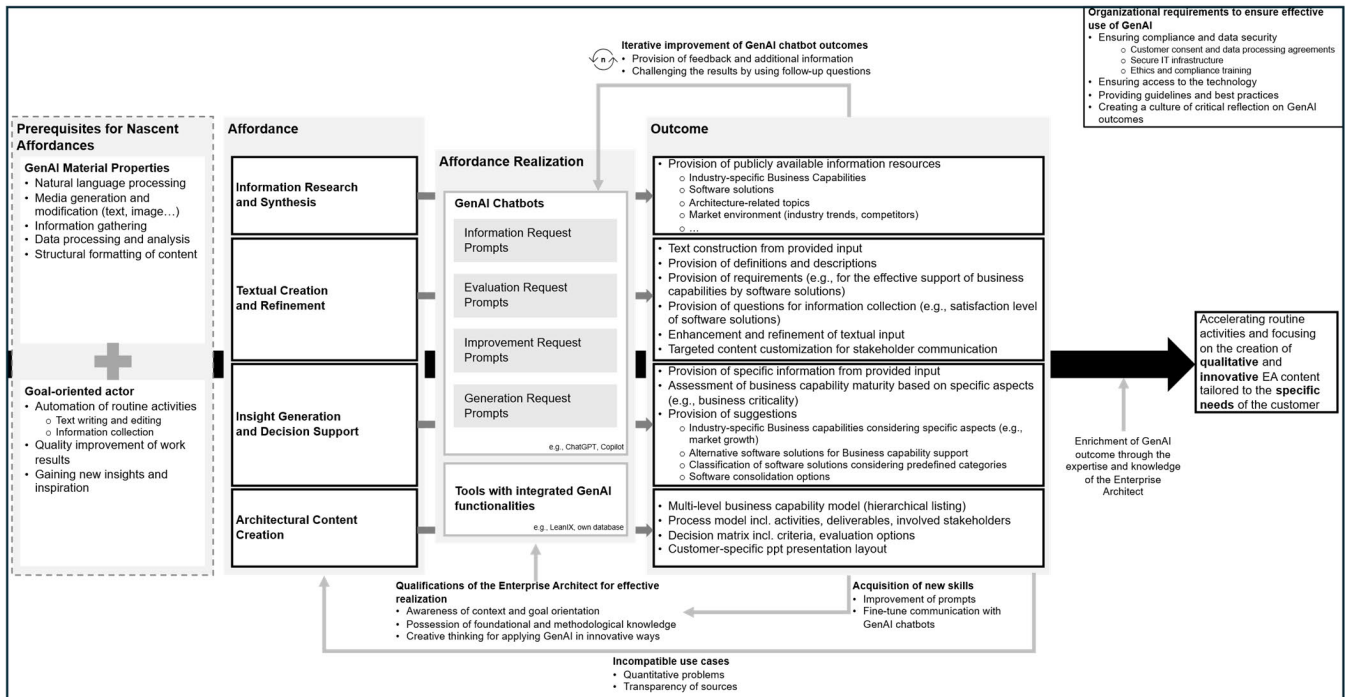


FIGURE 1. A framework of the affordances provided by generative AI in the context of enterprise architecture.

inductively. Finally, the data served as the foundation for constructing the framework of the affordance actualization process by GenAI in EA.

IV. RESULTS

We identified different GenAI material properties and specific enterprise architects' goals as prerequisites for the emergence of nascent affordances. Moreover, we derived four key affordances and the mechanisms to realize those affordances to achieve EA's desired outcomes. Finally, the necessary organizational requirements and the required qualifications of enterprise architects were elicited that enabled the successful implementation. Figure 1 provides an overview of the resulting framework for the affordances of GenAI in the context of EA.

A. PREREQUISITES FOR NASCENT AFFORDANCES

The affordances of GenAI in EA arise from the combination of GenAI's technical functionalities (i.e., its material properties) and the motifs of the enterprise architects to integrate these technologies into their activities.

1) GenAI MATERIAL PROPERTIES

Natural language processing (NLP) enables GenAI systems to comprehend and process text-based input. This allows enterprise architects to provide AI systems with precise instructions that enable them to generate coherent and specific responses. One enterprise architect describes the functionality of NLP as follows: "I see GenAI as an assistant. Interacting with this assistant is like interacting with a

colleague through language, except that GenAI is all about prompts. And I would say that the basic function that GenAI offers is the understanding of human language. Since GenAI is able to understand me, I can use it effectively because it understands my requests and can then implement them. It's not perfect, but the basic understanding is there. And if it doesn't understand me, I can also give feedback so that ChatGPT, for example, is steered in the right direction." (Respondent 02)

Media generation and modification refers to the ability of GenAI systems to create and modify content in a variety of modalities. In the context of text modalities, the functionality enables generating text-based content, ranging from the simple rephrasing of a sentence to the creation of comprehensive and detailed texts. The enterprise architect can leverage this functionality to translate thoughts and concepts into clear and understandable texts or utilize GenAI to create texts that take specific criteria and contextual information into account when generating text. In the context of the image modality, this enables creating visualizations such as images or graphics. Respondent 02 provides an example of how he could use the image generation feature of GenAI for EA: "But now with DALL-E within ChatGPT Pro it works quite well that you can display specific images, graphics, visualizations, concepts. So, if I say, for example, illustrate the concept of security, or something like that, then it generates a lock for me." (Respondent 02)

Information gathering allows access to information on specific topics and is therefore often compared to a search engine. This function simplifies the process of gathering

information from many sources for enterprise architects, allowing them to develop a basic understanding of new, unknown topics. This also allows the enterprise architect to respond expeditiously to evolving customer demands, aiding in the development of solutions. Respondent 01 explains how he sees the information gathering property similar to a search engine but more efficiently: “[I]f you make a prompt and expect a text as a response and [...] then you use ChatGPT like an additional front-end in the sense of a search engine. It is less generative in terms of creativity but is an accelerator that makes the research much faster than I am able to do myself.” (Respondent 01)

Next, **data processing and analysis** is an important GenAI property for EA as it enables to gain customized insights or obtain recommendations through GenAI systems based on provided data. Respondent 04 describes the functionality as follows: “Where I see potential added value is in the analysis of my data. I’m not specifically talking about customer data here, because of course we’re not allowed to give that to ChatGPT. But a specific prompt is actually data too, that can have more content or be described in very general terms. Then I can ask ChatGPT for its opinion on the content. It has to analyze the content in order to give me an output in response to my request.” (Respondent 04)

The final GenAI material property involves the **structural formatting of content**. This enables enterprise architects to not only create textual content in a variety of structured formats – such as lists, notes, or tables – but also convert texts from one format to another. As a result, enterprise architects can enhance the clarity and comprehensibility of information. Respondent 02 provides an example to illustrate the functionality of structural formatting of content: “If I have a continuous text with several points, I say, okay, let’s summarize this in a table or let’s make a comparison. Or the other way around, if I say, here we have the comparison, then I say, maybe create a summary. [...] I give my mostly custom GPT instructions, such as I just want this format, give me this and that, pay attention to the text length and the language and so on.” (Respondent 02)

2) GOAL-ORIENTED ACTORS

In terms of enterprise architects’ motivations for integrating GenAI into their work, the primary drivers are the automation of routine tasks, the independent improvement of work quality, and the creation of new insights and inspiration.

Automating routine activities refers to activities of writing and editing text and gathering information, which were common themes by the experts that are supported by GenAI use. Automation accelerates manual routine activities, which is fundamental for increasing the efficiency of enterprise architects and is valuable in day-to-day project work with limited time resources. By freeing themselves from routine activities, enterprise architects can focus on more challenging and higher value activities, such as value-added customer interactions. Respondent 07 describes the benefit of automating routine activities as follows: “So, GenAI simply helps with

efficiency. While you always have to check the quality again, it’s simply a helper and accelerator for most of my routine work.” (Respondent 07)

The use of GenAI enables the automation of the text creation process, reducing the effort required for information compilation and text production, as respondent 02 describes it: “And what I find extremely exciting about [GenAI] is the way the answers are formulated. And that’s very different from the way it used to be when I still had to somehow formulate a sentence or a paragraph from provided data. Maybe even translate from different source languages. That’s all been removed now with these new tools.” (Respondent 02)

GenAI not only accelerates the collection of information but also fosters the diversity and comprehensiveness of information on a topic or area of work that would otherwise be restricted due to time constraints. Respondent 01 explains in his view how GenAI contributes to the completeness of information, especially given the limited time available in day-to-day project work: “Time constraints play a big role, because I don’t have unlimited time for research in my day-to-day work and whatever I find in that time becomes the basis for my analysis. That always carries the risk of relying on outdated or incomplete information or overlooking something important. But GenAI helps me be both faster and more thorough within the limited time I have.” (Respondent 01)

Another rationale for enterprise architects to integrate GenAI into their daily operations is to **improve the quality of their work**. By obtaining initial feedback from GenAI, they can enhance the quality of their work without having to depend on the availability of colleagues or employees, as respondent 02 explains it: “I can talk to other people about things like that, but people are also busy and then it just helps if you ask ChatGPT and get feedback and then you can also save time. You can do it on your own, you don’t have to approach others, you don’t have to annoy others. And AI often has more knowledge than the random colleague who is not specialized in my topic.” (Respondent 02)

Moreover, GenAI can be employed to critically examine textual content or viewpoints on a topic, ensuring that all relevant aspects have been sufficiently considered and that assumptions are confirmed externally. In particular, the quality of the work can be improved iteratively by reusing proven prompts to GenAI systems with contextual content adjustments. Respondent 06 outlines how he can use GenAI to ensure that his own opinion or initial thoughts are reflected in the final documents, contributing to a higher quality standard: “I believe, especially when it comes to challenging something again, that you really do have a certain level of quality assurance. So, when it comes to texts or things that you might have an opinion on, that you can have it challenged again independently by AI. I think it definitely helps to improve quality. Or to make sure that certain things have perhaps not been left out.” (Respondent 06)

Another rationale for enterprise architects to employ GenAI is to **obtain new insights and inspiration**. GenAI can propose approaches or methodologies in domains that

may be unfamiliar, thereby facilitating the starting phase of projects and enabling deeper engagement with the subject matter. It is also capable of generating ideas that lie beyond the architect's current perspective or exceed their area of expertise and conventional modes of thought. Respondent 10 explains how he uses GenAI as an enterprise architect to gain inspiration and help him to build on those ideas: *"It's like having a brainstorming assistant. When I'm assigned a new topic that I'm not very familiar with, I use GenAI to quickly get a basic understanding—just the key points, what the topic is about, and how I might approach the problem. I don't use the AI's output directly in customer interactions, but it helps me get up to speed faster than I could on my own. From there, I build on that input to develop my own ideas and find a solution."* (Respondent 10)

Finally, the evaluation of data input from AI can reveal novel aspects, patterns, and properties of which the enterprise architect was previously unaware. Respondent 05 describes how GenAI has given him new insights that he had not previously considered or conceived: *"[You] are running some analysis or something and you check with the GPT and suddenly the GPT mentions some of the aspects that you didn't consider before. And sometimes you say no it's not what I mean but most of the time it's pretty good."* (Respondent 05)

B. GenAI AFFORDANCES FOR ENTERPRISE ARCHITECTURE

Regarding the affordances of GenAI in EA, Table 2 gives an overview of the identified prerequisites from the previous section that enable specific nascent affordances.

TABLE 2. Overview of prerequisites for nascent affordances in the context of EA.

Material Properties	Goal-oriented Actor	Affordance
- Natural language processing - Media generation and modification - Information gathering	Automation of routine activities	Information Research and Synthesis
- Natural language processing - Media generation and modification - Structural formatting of content	- Automation of routine activities - Quality improvement of work results	Textual Creation and Refinement
- Natural language processing - Media generation and modification - Data processing and analysis	- Quality improvement of work results - Gaining new insights and inspiration	Insight Generation and Decision Support
- Natural language processing - Media generation and modification	- Automation of routine activities - Gaining new insights and inspiration	Architecture Content Creation

The **"information research and synthesis"** affordance results from the material properties NLP, media generation and modification, and information gathering, as well as the intent of the enterprise architect to automate his routine activities. It encompasses the support of enterprise architects in identifying, collecting, and synthesizing relevant information and data sources. Respondent 01 describes the potential support that GenAI provides in the collection of information: *"[M]y work as an architect is often associated with repetitive work and research. [...] I'm still relatively slow at the moment because I'm somehow dependent on finding the right information which sometimes is hard to find. And GenAI, ChatGPT, helps me a lot in synthesizing the right information."* (Respondent 01)

Similarly, Respondent 04 states that GenAI assists him in gathering information, particularly in subject areas with which he is less familiar: *"At the moment, GenAI is more of a shortcut for research activities for me. Every time I'm researching something, I would just use it to speed things up. I would also use it to perhaps look up topics that I'm not particularly familiar with, because the information that I would otherwise have to manually pull out of the website is easier to find."* (Respondent 04)

The data gathered by GenAI is sourced from publicly available online resources. The primary objective of GenAI is to retrieve this information automatically, thereby eliminating the need for manual research by enterprise architects. Respondent 01 describes the affordance of information research and synthesis as follows: *"And there are many other use cases I could tell you about, but they all have one thing in common: the knowledge about it already exists on the Internet and is becoming more and more. You just have to find it, collect it, categorize it, sort it and somehow make a selection. In other words, perfect use cases for GenAI."* (Respondent 01)

GenAI not only supports information collection and makes it available to enterprise architects but also structures this information in a manner that facilitates its comprehension and clarity. Respondent 09 describes the structured information presented using GenAI: *"Like you said, research. You can simply give a prompt and then you'll get a certain structure. It's very impressive how structured the answers are."* (Respondent 09)

The **"textual generation and refinement"** affordance is the result of the material properties, including NLP, media generation and modification, as well as structural formatting of content, and the intention of the enterprise architect to accelerate his routine tasks and to improve the quality of his work through the assistance of GenAI. It enables enterprise architects to utilize GenAI to generate textual content and revise existing texts and adapt them to improve content clarity.

For instance, enterprise architects employ the **"text generation and refinement"** affordance to generate descriptions of specific artifacts. Here, GenAI can be employed in two distinct modes: in the initial approach, the AI model is

tasked with the generation of texts or descriptions without the specification of the content (i.e., zero-shot prompting). In the alternative case, a general outline of the content to be described is given, which the model then utilizes to create the desired descriptions with guidance. Respondent 06 explains how GenAI supports him in text generation, using the example of defining specific business capabilities: *“Text generation can be useful in many areas. For example, when we’re developing capability models or reference frameworks that need to be tailored to specific customer requirements. Since these capabilities also need clear definitions, GenAI can be a big help. You can simply say, ‘Here’s scenario XY—this area is very important for the company—now give me a few relevant capabilities and define them.’”* (Respondent 06)

As mentioned, GenAI can not only generate text without content input but also based on existing content. Respondent 02 describes how GenAI helped him to generate descriptions based on keywords: *“Enterprise architects often talk about business capability maps. [...] A company has 100 capabilities, which contain things like executing marketing campaigns and answering customer relationship support requests. [...] And then you have a catalogue, and these are always just keywords. I can just take ChatGPT and say okay, complete this catalogue with capabilities and give me a description of them.”* (Respondent 02)

Hence, this affordance not only enables the generation of textual content but also its refinement for enterprise architects. This serves to formulate the textual input more precisely or tailor the content to specific target groups in terms of the level of detail. Respondent 10 highlights how GenAI has allowed him to adapt existing texts in terms of style and language to better suit a particular audience or context: *“And I think that’s where AI can help, because even today we already have use cases where people say, okay, please formulate this paragraph in a more child-friendly way or formulate it in a more professional way.”* (Respondent 10)

The **“insight and decision support”** affordance results from the material properties NLP, media generation and modification, data processing and analysis, as well as the intention of the enterprise architect to improve the quality of his work and gain deeper insights into their provided data. GenAI primarily processes textual inputs or documents by analyzing and interpreting them. The results of these analyses provide enterprise architects with the necessary information to make informed decisions. Enterprise architects essentially use this affordance to analyze business documentations. Based on specific queries, they can leverage GenAI to summarize and aggregate the essential core information. Respondent 08 explains how he synthesizes large documents on specific topics of interest: *“System documentation is also highly relevant, especially in the context of solution architecture. While it may only be of secondary or peripheral interest to enterprise architects, depending on the case, the information it contains can still be very valuable. Why wouldn’t you use that today, because it’s simply uneconomical to look it all up? The problem is that manually going through all of it is*

often too time-consuming to be practical. But with GenAI, you can have it read the documentation and extract the key information for you.” (Respondent 08)

In addition to summarizing and extracting documents, GenAI can also perform more in-depth analyses. This enables the enterprise architect to integrate GenAI into tasks that would normally require their own personal decision-making skills. Respondent 05 uses the example of application classification to describe how GenAI can perform tasks independently by analyzing context: *“I provided a brief description of what constitutes a support application, a business application, and an innovative application, which include those built on new technologies. Based on these definitions, ChatGPT was able to classify various applications into the appropriate categories.”* (Respondent 05)

Furthermore, the enterprise architect can also use in-depth analysis from GenAI to generate suggestions that support them in their decision-making, enabling them to take data-driven actions of which they were previously unaware. Respondent 07 describes how GenAI assisted him by analyzing the data provided and independently suggesting areas for improvement: *“Certainly also to identify optimization potential, for example in the information architecture layer, when it comes to data flows and APIs, as the number is increasing exponentially, especially with new technological trends such as microservices etc., then to analyze current data flows etc., but not only to analyze them, but then also to generate proposals for optimization and consolidation. In the direction of an API data strategy.”* (Respondent 07)

Last, the **“architectural content creation”** affordance arises from the integration of GenAI material properties, including NLP, media generation, and modification, as well as the enterprise architect’s intention to accelerate routine activities and gain new insights or inspiration. It refers to the use of GenAI by enterprise architects to generate conceptual, architecture-specific planning and decision support tools. Respondent 09 illustrated how he instructed ChatGPT as a GenAI chatbot to generate a process model for a capability map. In addition, the affordance enables generating material tools that enterprise architects require for their tasks. Respondent 06 describes how he used GenAI to create a PowerPoint deck that was specifically tailored to the client’s industry: *“We also have an internal GenAI tool that can, for example, generate a first draft of a PowerPoint deck, including images, for a specific topic.”* (Respondent 06)

C. AFFORDANCE REALIZATION

Overall, we identified two options for implementing the affordances offered by GenAI in EA: through a distinct GenAI chatbot or through tools with integrated GenAI capabilities. Some affordances can be realized using only one of these options, while others can be realized using both options. Table 3 provides an overview of the possibilities in which the affordances provided are realized by enterprise architects. It shows the used tools and how they are employed

TABLE 3. Overview of the affordance realization possibilities.

Affordance	Affordance Realization		Example
	Used Tools	Actions	
Information Research and Synthesis	Generative AI Chatbot	Information Request Prompts	The enterprise architect uses a text-based GenAI application to prompt for the essential business capabilities that companies in Industry X need to be successful.
Textual Creation and Refinement	Generative AI Chatbot	Optimization Request Prompts	The enterprise architect uses a text-based GenAI application to tailor communication content to the specific needs and comprehension levels of different audiences, using a prompt to maximize message clarity and effectiveness.
	Generative AI Chatbot	Generation Request Prompts	The enterprise architect uses a text-based GenAI application to suggest definitions/descriptions for the business capabilities using a prompt.
	LeanIX	Use of the provided functions	The enterprise architect uses the description function within the LeanIX enterprise architecture tool to automatically generate capability descriptions.
	Microsoft Excel	Use of the provided functions	The enterprise architect uses specific prompts within an Excel list connected to OpenAI via an API to retrieve application descriptions for the application names listed.
Insight Generation and Decision Support	Generative AI Chatbot	Evaluation Request Prompts	The enterprise architect uses a text-based GenAI application to request an assessment of the business capability's maturity using a prompt based on predefined aspects such as business criticality (24/7 availability), market differentiation, and business support maturity with a predefined range of scores.

TABLE 3. (Continued.) Overview of the affordance realization possibilities.

Generative AI Chatbot	Generation Request Prompts	The enterprise architect uses a text-based GenAI application to prompt a list of applications that have a discrepancy of at least two maturity levels between the business capability and the supporting application, providing information on the current maturity level of application support and the desired maturity level to optimally support the business capability.
Generative AI Chatbot	Information Request Prompts	The enterprise architect uses a text-based GenAI application to identify the most important business capabilities for the company located in a specific industry, taking into account certain aspects such as market growth, by using a prompt.
Generative AI Chatbot	Optimization Request Prompts	The enterprise architect uses a text-based GenAI application to deliver recommendations tailored to specific customer characteristics, such as industry affiliation and company size, using a prompt to make more relevant and customized suggestions.
Microsoft Excel	Use of the provided functions	The enterprise architect uses specific prompts within an Excel list connected to OpenAI via an API to classify applications into specific categories. He also provides the classification descriptions to specify the desired classification.
Data Management System	Use of the provided functions	The enterprise architect uses a database equipped with GenAI capabilities to automatically generate optimization suggestions from the data provided (such as patterns and dependencies in data streams that indicate missing integrations between applications).

TABLE 3. (Continued.) Overview of the affordance realization possibilities.

Architectural Content Generation	Generative AI Chatbot	Generation Request Prompts	The enterprise architect uses a text-based GenAI application to create a topic-specific PowerPoint layout deck from a prompt that covers the customer project in terms of layout and topic.
	Custom Developed Application	Use of the provided functions	The enterprise architect uses the business capability model generator to automatically generate a graphical business capability map.

to implement these affordances, accompanied by illustrative examples.

The interviewed enterprise architects employ a range of solutions including OpenAI’s ChatGPT, Microsoft’s Copilot, and an internally developed chatbot from the case company to interact with **GenAI chatbots**. The affordances are primarily achieved through the execution of specific actions in the form of natural language prompts provided by the users. These actions can be classified into four distinct prompt types: information request prompts, optimization request prompts, evaluation request prompts, and generation request prompts.

- **Information request prompts** are utilized by enterprise architects to instruct the GenAI chatbot to provide core information on specific topics. Respondent 03 uses a use case to illustrate how he instructed GenAI chatbots to provide him with information on a specific topic using an information request product: “If you want to quickly get some product names for any technologies or special areas, I’ll ask ChatGPT to give me the information so that I can then use the input to move on.” (Respondent 03)
- With **optimization request prompts**, the enterprise architect instructs GenAI chatbots to enhance the quality of text content or identify potential areas for improvement. Respondent 03 uses an example to illustrate how optimization suggestions can be generated using prompts in GenAI chatbots: “[I] can also say, for example, the customer is in industry X and has size XYZ, what would you adapt? And then I can also ask, okay, now we have three things, what other alternatives can you think of, what would you recommend? That’s how I would use it, if I have a specific question, that I simply pass it on to ChatGPT.” (Respondent 02)
- **Evaluation request prompts** are used to instruct the GenAI chatbot to provide evaluations of states or performances based on predefined criteria, as Respondent 01 outlines: “This evaluation aspect can also be

included in a prompt and works relatively well. There are almost no limits to the prompts.” (Respondent 01)

- **Generation request prompts** instruct GenAI chatbots to create new textual, conceptual, or material content based on a specific context. Respondent 01 explains how he uses GenAI chatbots to create conceptual models: “For example, I instruct ChatGPT to generate a process model for a capability map.” (Respondent 01)

In addition to GenAI chatbots, enterprise architects are also using **tools with GenAI capabilities**. These tools are either already equipped with GenAI capabilities by their providers or equipped with generative capabilities by connecting to services such as OpenAI’s LLMs via API connections. Among the reported tools used are the EA tool LeanIX, Microsoft Excel, various database systems and custom developed applications.

To actualize the “**information research and synthesis**” affordance, enterprise architects primarily employ GenAI chatbots and use information request prompts to gain structured core information on the desired topics. For instance, enterprise architects can engage with GenAI to research common capabilities and the associated definitions that are typical for companies in a particular industry of their clients: “We architects are often concerned with capability maps. For these, you can also generate a textual representation of a capability map which usually consists of various titles that each describe the business capability and an explanation to describe it in more detail. So, the content matters. And these results can be wonderfully generalized, for example, across industries or across companies, industry-specific. You can say that a company in a certain industry has a very high probability of having 90 to 95 percent of the same capabilities. And this knowledge is available on the Internet. You just have to collect it and categorize it. [...] This is of course the perfect use case for GenAI in my opinion.” (Respondent 01)

In the context of business capabilities, enterprise architects can also use GenAI chatbots to enact the “information research and synthesis” affordance and suggest common software solutions that enable the technical implementation of the respective capability in the company. Respondent 08 describes how he prompts a GenAI chatbot to obtain information about software solutions for technological support of the capabilities: “You can just ask about the common software solutions in industry XYZ for company ABC that are used for these capabilities. [...] What capabilities should such a company have? What does it need and how is it also technologically enabled? In the industry overview for this example, this [ChatGPT] also provides you with a pretty good idea.” (Respondent 08)

In order to actualize the “**textual creation and refinement**” affordance, enterprise architects primarily leverage GenAI chatbots, but they also employ tools in certain use cases that offer dedicated generative functions to draft entire documents like requirements specifications or questionnaires, often reducing hours of manual work. The

realization of this affordance can result in three different outcomes: First, texts can be generated based on provided content; second, existing texts can be improved in terms of their textual quality; and third, content can be adapted to specific interest groups.

For instance, Respondent 08 shares how he prompted an initial requirements document tailored to a specific capability and solution, highlighting GenAI's effectiveness in producing practical content. In addition, Respondent 04 describes generating descriptive texts for business capabilities quickly and accurately, which otherwise would have required extensive research or reliance on legacy material: *"This is a typical task that would normally require a lot of research, like writing a descriptive text. I would have either spent much more time thinking it through or just pulled information from internal documents—probably buried somewhere in old PowerPoint slides. But with ChatGPT, it was much easier. I just entered my prompt, and the response was actually quite accurate and comprehensive."* (Respondent 04). Similarly, Respondent 2 reported an example of how he used the EA tool LeanIX to obtain pre-formulated definitions of business capabilities that can be directly used in the next work steps.

Moreover, enterprise architects use GenAI not only to create text but also to polish and optimize existing documents. Respondent 03 explains: *"I use ChatGPT to optimize formulations for core artifacts such as architecture principles... It actually does that very well."* Thus, GenAI enables the improvement of grammar, spelling, wording, and phrasing. Especially non-native English speakers benefit from the natural language capabilities through optimization request prompts for textual refinement.

Another valuable benefit is adapting texts for various audiences such as stakeholders to make communications more relevant to business stakeholders or IT professionals without extra effort: *"From my point of view, one advantage of GenAI beyond simply generating texts is the ability to support a kind of 'multilingualism' in the sense of adapting tone and style for different audiences. For example, it can easily create content with a more formal, business-oriented tone when needed, or make it more specific and technically precise for IT professionals. And it can do this without requiring much additional effort."* (Respondent 08).

To realize the **"insight generation and decision support"** affordance through GenAI, enterprise architects mainly rely on GenAI chatbots, but they are also equipped with tools that have built-in GenAI capabilities. When implementing this affordance, three specific outcomes can be achieved: the provision of specific core information from the given input, the evaluation of an object based on specific aspects, and the creation of suggestions by GenAI. Enterprise architects employ it to extract and summarize essential information from large amounts of documents, such as Request for Proposals (RFPs), enabling faster comprehension of key points (Respondent 06). Information request prompts are leveraged to specify which goals and specific information they would like to receive from the provided input. GenAI is also employed for

evaluation purposes, such as to evaluate organizational capabilities and their maturity levels by conducting in-depth processing and analysis of data with evaluation request prompts. For instance, Respondent 01 explains the process for comparing current states of a customer's applications with target architectures that facilitates workshop preparation and customer dialogues: *"So, part of the work involves evaluating maps: capability maps, support maps, and so on. Typically, you do this in an 'as-is' and 'to-be' comparison with the customer to identify gaps or discrepancies. Once those gaps are identified, you can ask GPT to, for example, generate a list of all cases where there's a discrepancy of at least two maturity levels between a capability and its supporting application. This helps to prepare this in advance for a workshop, providing an 'outside-in' perspective and allows us to go straight into detailed discussions with the customer, without having to build the entire assessment from scratch during the session"* (Respondent 01). Furthermore, GenAI proposes suggestions, including brainstorming additional business capabilities or identifying alternative software solutions based on specific criteria that traditionally required extensive manual research (Respondent 02, Respondent 08). Here, GenAI's capacity to automate lengthy processes such as to categorize and classify thousands of applications stands out: *"I provided short descriptions of what a support application is, what a business application is, and what we mean by an innovative application: specifically, applications that are new and based on emerging technologies. Using these definitions, GenAI was able to classify the listed applications into these three categories. I remember doing this classification via the GPT API. I set it up directly from Excel, launched the process, and left it running overnight. The next morning—boom—I had the classification results for seven thousand applications."* (Respondent 05). Enterprise architects even use AI-generated recommendations for consolidating software portfolios, streamlining complex landscape rationalization (Respondent 10, Respondent 07).

Finally, enterprise architects utilize GenAI chatbots as well as GenAI-enabled EA tools to realize the **"architectural content creation"** affordance for aiding the generation of conceptual models and presentation materials. By using generation request prompts, they leverage GenAI to outline process models or develop capability frameworks, which serve as foundational guides for whole workstreams. Respondents described how GenAI quickly generated structured activities and phases, enabling rapid progress and creative inspiration: *"I define the subject area, the direction I want to take, and any constraints I'm working with. Then I ask ChatGPT to suggest five phases, each with three activities and three outputs that would help me reach my goal in that context. In some cases, the results not only came incredibly fast, but they also provided really useful input which I either wrote down directly or used as inspiration for further work."* (Respondent 01). Moreover, multiple enterprise architects reported that GenAI can generate hierarchical business capability models across domains and subdomains, assisting in industry

benchmarking and internal development. Some organizations have even advanced this by creating custom GenAI tools that automatically produce graphical capability maps based on multimodal capabilities of LLMs (Respondent 01). Additionally, GenAI helps generate initial decision matrices that compare options on multiple criteria, facilitating structured decision-making (Respondent 02). Lastly, enterprise architects also employ GenAI in supporting material preparation, such as drafting PowerPoint presentations complete with images for client communication by specifying the desired look and layout using generation request prompts: *“We also have an internal GenAI tool that can automatically generate a first draft of a PowerPoint deck, including images, for a specific topic that you specify with an instruction.”* (Respondent 06)

D. AFFORDANCE OUTCOMES

The outcomes generated by GenAI currently serve as a foundation for enterprise architects, who **enrich them with their knowledge and experience** to be able to utilize them in customer projects. Respondent 04 provides an insight into the current state of direct adoption of GenAI results, stating that *“it is probably still problematic at the moment to adopt the results one-to-one”* (Respondent 04).

This results from the missing assurance that AI-generated results are fit for purpose and accurate. For instance, Respondent 05 asserts that the user retains ultimate control, regardless of the level of GenAI use. Additionally, Respondent 01 outlines how GenAI provides suggestions for business capabilities by initially generating a semi-finished version for a specific industry. The time saved allows him to immediately start quality control procedures and further improvements. By contributing his expertise, he transforms the initial output of GenAI into a fully developed product: *“It used to take us around two weeks to get this done. Now, we can generate an 80% version of the business capability model in just three minutes. Of course, it’s not as good as the models we build ourselves, but the key advantage is that we can start improving and quality-checking it right away. Before, I would have to wait two weeks for my analyst to deliver the initial version before I could even begin making improvements.”* (Respondent 01)

Yet, respondents discussed that if the effectiveness of using GenAI depends on enterprise architects using their expertise and experience to create practical results from the initial generated outputs, this implies that junior enterprise architects with less experience or knowledge may be less able to use AI as effectively as their more experienced colleagues. For instance, Respondent 06 offered the following comment: *“Yes, I would definitely say that. I also think it takes a certain level of experience to use GenAI effectively. If you’re working without much background or context, the results might not be very useful. But if you can combine the AI’s output with your own experience... things you’ve seen before or a sense you’ve developed from working with a customer, then you can achieve much better results.”* (Respondent 06)

The enterprise architects were able to recognize the new skills that they have acquired **to achieve higher quality outcomes through the use of GenAI**, resulting from the implementation of affordances and the resulting outcomes. One skill that enterprise architects have developed or improved using GenAI is the **optimization of prompts**. When using GenAI systems, enterprise architects have found that the quality of the results significantly depends on the design of their prompts. To achieve more comprehensive results, they have learned to formulate the problems and framework conditions to be solved more precisely in their prompts. This enables GenAI to support them more effectively. Respondent 03 elucidates this as follows: *“Yes, I think you have to approach it differently. You need to ask yourself: what is the exact question I’m trying to answer? Instead of jumping straight to the solution, you’re forced to formulate a precise prompt, which means you have to clearly define the question. In doing so, you also start to consider certain framework conditions that you might not normally write down explicitly, but that you usually just keep in mind.”* (Respondent 03). Furthermore, Respondent 08 was able to notice that the more frequent use of GenAI led to an incremental improvement in their prompts, ultimately resulting in the desired outcome of higher quality.

Another skill that enterprise architects have acquired in dealing with GenAI is the ability to **fine-tune communication with GenAI chatbots**. By using GenAI, they have learned what information and requests they need to provide and how the AI system reacts to certain formulations. One respondent describes his experience as follows: *“Exactly. You start experimenting—defining GenAI’s role, the context, and expected outputs. It takes practice to consider all these aspects and to learn how GenAI responds to your wording. Over time, I’ve noticed I’ve improved in asking better, more precise follow-up questions without losing context.”* (Respondent 08)

In addition to developing and enhancing capabilities, the enterprise architects were also able to identify areas of application in which GenAI is currently unable to provide effective support. They discovered that the outcomes produced by GenAI were not suitable for **quantitative problems**. For instance, GenAI chatbots provided unreliable results in quantitative use cases for the enterprise architects that could be due to hallucinations: *“But whenever it’s really about numbers—things like benchmarks, cost savings as a percentage of revenue, license costs for specific software, or estimated project durations—ChatGPT always starts by giving a long explanation about how these figures are highly customer-specific and need to be analyzed in detail. But then, it often starts to hallucinate and produces numbers that aren’t grounded in reality and have no real connection to the actual situation.”* (Respondent 01)

Finally, enterprise architects have identified issues with the **transparency of sources**, which emerged when they attempted to view the sources used by GenAI chatbots for the purpose of verifying results or conducting further research.

Respondent 02 describes his experience with source transparency as follows: *“Sometimes I get answers like ‘Here’s a list of the ten best CRM tools,’ but when I check, they don’t actually fit. Then I ask again, and the same tools keep coming up. Sometimes it even provides links to websites that don’t exist, or it lists tools that aren’t real or link to completely unrelated pages. The source citations also don’t work very reliably. You often run into this issue...if you have certain expectations, ChatGPT frequently doesn’t fully meet them. You can always refine your prompt to improve the answer, but sometimes it just keeps repeating the same things.”* (Respondent 02)

E. ORGANIZATIONAL CONTEXT AND QUALIFICATIONS

For GenAI to be effectively utilized within the **organizational context** of EA, certain requirements must be met, which can be broadly classified into four main categories: ensuring compliance and data security, providing access to GenAI technologies, offering guidelines and best practices, and fostering a culture of critical reflection on the results of GenAI.

In order to **ensure compliance and data security** while employing GenAI, companies must comply with relevant regulatory requirements and maintain a robust and secure information technology infrastructure, particularly when utilizing GenAI in the context of customer data. As Respondent 06 notes, large-scale deployments for customer projects demand *“an overall safeguard, [...] legal sign-offs, and perhaps even an additional agreement or coordination with individual customers and individual projects that GenAI may be used”*. Moreover, companies must provide their users with secure, compliant environments to interact with GenAI. Respondent 05 points out that while it is currently *“left to everyone to keep the security compliance”*, this approach carries risks, and Respondent 01 adds that users need adequate training to recognize when their prompts might violate copyright infringements or NDAs because even more experienced GenAI users struggle with determining the boundaries of compliant use.

Mandatory training is critical to ensure ethical and safe use of GenAI. As Respondent 09 remarks, employees need *“the right sense of proportion to know that a certain GenAI use is still within the legal framework”* and this should become part of their extended skillset. Secure environments must also prevent data leaks, including ensuring customer data is not used to train public AI models. Respondent 07 is clear: *“Compliance must be ensured so no information is passed on [...] and not used for training purposes either.”* Implementing enterprise-grade GenAI solutions such as Microsoft Copilot with GDPR compliance or local GenAI systems can help guarantee data does not leave company boundaries and that internal materials are only used appropriately (Respondent 06).

In terms of providing **access to technology**, organizations should ensure all staff can use up-to-date and secure GenAI solutions. However, as Respondent 08 observes, *“access to*

these tools is not yet completely barrier-free [...] and the necessary licenses are not yet available for everyone.” Neglecting to provide current tools could lead users to rely on outdated or non-compliant alternatives, risking further compliance breaches (Respondent 02).

Providing practical guidelines and best practices are essential, particularly for inexperienced users. Respondent 06 emphasizes the value of a documented library of use cases and best practices, allowing employees to *“not reinvent the wheel every time”* and work productively by facilitating sharing and reusing proven solutions. Guidelines should also address prompt creation and compliance, since, as Respondent 05 explains, many employees *“just don’t know how to make it better”* and need clear instructions to avoid exposing internal data.

Finally, **fostering a culture of critical reflection** means encouraging enterprise architects to always critically question and scrutinize GenAI outputs. Respondent 01 underscores that *“lazy thinking is an absolute killer as a strategy and enterprise architect,”* and that the increasing sophistication of AI tools within everyday work only raises the bar for thoughtful and critical engagement: *“[...] Ultimately, I mean ChatGPT, Generative AI is simply the next tool...And I can be an artist with all tools or a monkey who presses the button. [...] But it’s true that the more sophisticated the tools become, the more demanding the use of the tool. And I think you have to invest in this ability to operate the tool. And with ChatGPT, this ability means thinking and differentiating.”* (Respondent 01). Only by investing in critical thinking can organizations ensure that GenAI serves as a tool for innovation, not just automation.

Enterprise architects should furthermore possess specific **qualifications** that enable them to actualize the affordances fully and leverage GenAI effectively in the context of EA. Three skills are especially crucial: context awareness and goal orientation, robust foundational and methodological knowledge, and creative thinking. As AI-generated results are of a probabilistic nature and thus rarely definitive, enterprise architects also need strong critical thinking and questioning abilities to iteratively refine outcomes, seeing GenAI as a collaborative partner in improving and perfecting the results.

The concept of **“awareness of context and goal orientation”** implies understanding the enterprise architect’s role within the context of the customer’s organization, the identification of its problems, and the ability to conceptualize solutions. It means understanding the unique circumstances and strategic objectives of the customer’s organization so GenAI can be employed as an effective support tool. Respondent 02 elucidates the significance of customer comprehension as follows: *“You have to understand the stakeholders in order to deliver good results. And if you don’t really understand the stakeholders, then AI is simply useless. Because then you will get things that don’t appeal to the stakeholders. And that’s not just specific to ChatGPT, but that’s where it becomes extremely important”* (Respondent 02). Respondent 03 adds that understanding the business strategy enables the

architect to craft more effective prompts that lead to better results: “Also the understanding of the business strategy, so in this prompt I also have to somehow say, what do I want to achieve? [...] These are advanced skills that no AI can do for you.” (Respondent 03)

Foundational and methodological knowledge is equally important. Enterprise architects should be able to ask the right questions and verify GenAI outputs based on their expertise. As Respondent 04 explains “[the enterprise architect] needs to have a solid understanding of their work and of the methodology, because GenAI is ultimately just a tool. They need to know how to ask the right questions, which requires a certain level of basic knowledge. That’s also what enables them to verify the information they get” (Respondent 03). Respondent 09 emphasizes the need for fundamental knowledge to spot limitations in AI-generated data, especially for tasks like calculations: “When you get a financial statement from ChatGPT, it’s really important to be able to calculate it back from the business report and understand it yourself.” (Respondent 09)

Creative thinking enables enterprise architects to apply GenAI in innovative and unconventional ways by demonstrating creativity and innovation. Respondent 04 shares an example of someone building an application with ChatGPT’s guidance without programming knowledge, highlighting the inventive uses possible: “You first have to come up with the idea of actually using ChatGPT for all these issues” (Respondent 04). Creative strategies are also emerging for verifying GenAI outputs. For instance, Respondent 01 describes sending the same prompt to GenAI multiple times, then tallying motifs that recur as a confidence measure to ensure the quality of industry-specific business capabilities provided by GenAI in one case: “By identifying which answers consistently appear, we can assign a kind of confidence score...we’ve partly automated the process of questioning and validating GenAI outputs” (Respondent 01). Ultimately, GenAI can act as an enabler and empowers its users in scenarios that were not previously considered due to the lack of knowledge or skills.

Finally, **critical thinking and the ability to question results** are necessary because GenAI’s initial outputs are rarely fully accurate or contextually relevant to meet quality standards. As Respondent 08 warns “there is a risk of combining information that is not quality-assured when using GenAI”. By subjecting the results to rigorous scrutiny, enterprise architects can ensure that they are of high quality. Respondent 05 emphasizes that a good enterprise architect is someone who “critically challenges the results generated by GenAI” and Respondent 02 adds that “most people would just take [the result] and present it, maybe they don’t even understand it. But a good enterprise architect should always be critical of the results of ChatGPT. He already has a vision and uses AI to supplement and challenge.” (Respondent 02)

Continuous refinement and adopting a critical attitude towards the results of GenAI is integral to optimizing its results. Respondent 04 advises: “Even more than with a

traditional search engine, it’s important to know how to ask the questions, how to use it, how to ask follow-up questions. You can keep refining the results...What you should also do is provide additional information after generating the first results. When I prompt certain things that ChatGPT is supposed to do, it has often happened to me that I didn’t provide all the information the first time.” (Respondent 04). The ability to iteratively improve results by specifying more context or asking better follow-up questions is a key competence for modern enterprise architects.

V. DISCUSSION

A. GENERATIVE AI AFFORDANCES IN THE CONTEXT OF EA

Based on interviews with ten EA experts, four key affordances of generative AI for enterprise architects were identified as well as their actualization process to enact those effectively. While these affordances offer significant potential benefits, our analysis reveals several critical tensions and limitations in their practical implementation.

The primary challenge lies in balancing efficiency gains with quality assurance requirements. Although GenAI can rapidly process information and generate content, enterprise architects must invest considerable time in verification and contextual adaptation. This verification burden is amplified by issues such as hallucination, outdated information, and the models’ tendency to generate superficial or decontextualized outputs. As noted by Dwivedi et al. [11], the quality of results fundamentally depends on training data quality, creating an inherent reliability risk.

A particularly significant constraint emerges from compliance and data protection requirements. Enterprise architects are often prohibited from sharing customer data with GenAI systems, limiting the tools’ ability to provide company-specific insights. This creates a practical barrier where GenAI outputs remain generic and require substantial human effort to contextualize for specific customer situations. The issue is further complicated by the “black box” nature of AI decision-making [82], making it difficult for enterprise architects to justify AI-generated recommendations to stakeholders.

The research also highlights limitations in GenAI’s capacity for empathy and innovation. While GenAI can effectively combine and repurpose existing knowledge, it struggles with truly novel solutions or understanding the human and political dimensions of organizational change. As noted by multiple respondents, this limitation is particularly evident in situations requiring careful stakeholder management or consideration of corporate politics. In terms of architectural content creation, while GenAI can assist with initial content generation, it currently serves primarily as a starting point rather than a complete solution. Enterprise architects must still translate AI-generated content into meaningful visualizations and ensure alignment with specific organizational contexts. This is particularly evident in business capability modeling, where industry-standard capabilities must be adapted to company-specific requirements.

Our findings suggest that the successful implementation of GenAI in enterprise architecture requires a carefully balanced approach. Organizations should view these tools as complementary to human expertise rather than replacements. This aligns with recent literature on human-AI collaboration [83], which emphasizes the importance of maintaining human oversight while leveraging AI capabilities.

Looking forward, several opportunities exist for improving the integration of GenAI in enterprise architecture. These include developing enterprise-specific AI models trained on proprietary data, implementing stronger privacy-preserving mechanisms, and creating more transparent AI decision-making processes. However, these solutions must address significant challenges around data protection, bias mitigation, and the need for explainable AI.

B. GENERATIVE AI AS MACHINE COLLABORATION PARTNER

Regarding RQ2, GenAI is increasingly utilized by enterprise architects as a **machine collaboration partner**, enabling the acceleration of routine tasks and facilitating creative content generation. This aligns with the concept of human-centered AI collaboration, wherein AI complements human expertise by providing initial insights, suggestions, and architecture-specific content [84].

Therefore, we derive five propositions for guiding enterprise architects in employing and using GenAI based on our results as well as subsequent comments from the interviews:

Proposition 1: Leverage GenAI tools and chatbots as collaborative partners to streamline routine content production and inspire innovation, while maintaining human expertise at the core of decision-making. Tools generate textual content but lack customization options, whereas chatbots enable refinement through iterative prompts. “*Chat-GPT spit it out...I'll take it that way. I've saved myself some work*” (Respondent 01) exemplifies potential risks of dependency. Thus, ensuring human oversight is crucial to preserving quality. Although GenAI accelerates information retrieval, it may undermine deep learning and independent thinking, impacting decision-making quality over time [85], [86]. Reliance on AI-generated conclusions without critical reflection may diminish innovation, as standardized solutions may not address unique challenges.

Proposition 2: Promote critical thinking by combining GenAI with structured learning approaches. Gamification and individual learning pathways [87] can reinforce foundational and advanced EA knowledge, counteracting knowledge superficiality.

Proposition 3: Use innovation-centric environments and collaborative challenges to mitigate reliance on GenAI's standard responses. Workshops fostering discourse, brainstorming, and trial-and-error methods encourage creative solutions.

GenAI also poses reproducibility and explainability issues due to its probabilistic nature [88]. Variability in outputs can lead to errors, compromising trust in AI-generated content. Over-reliance may result in overlooked inaccuracies [89].

Proposition 4: Establish feedback systems and version control to validate GenAI outputs [90]. Ensuring iterative model improvements reduces risks tied to hallucinations, where models produce misleading or fabricated information [91]. To mitigate hallucinations, robust data governance and data cleansing processes are critical [92]. Ensuring reliable training datasets minimizes factual inaccuracies.

Proposition 5: Invest in comprehensive data quality management and regular audits to enhance the reliability of GenAI outputs. Presenting both optimal and flawed case studies can heighten awareness of GenAI's limitations and need for verification.

In summary, GenAI significantly enhances efficiency and ideation in enterprise architecture but must be balanced with human expertise, critical review, and thoughtful integration. By anchoring AI as a supportive tool within robust educational, innovation, and quality assurance frameworks, organizations can harness their full potential while maintaining decision-making integrity.

C. RESEARCH IMPLICATIONS

The findings of our research contribute to multiple streams of technology adoption theory, particularly regarding generative AI [9], [61], [62], by demonstrating that the successful actualization of GenAI affordances in enterprise architecture is contingent not only on technological capabilities but also on complex sociotechnical conditions. Our study highlights that affordance actualization requires interplay between material properties of GenAI, user goals and expertise, and organizational factors such as governance maturity and data quality. This aligns with and extends existing frameworks that emphasize organizational readiness and dynamic capabilities as critical enablers of AI adoption and value realization in large enterprises [93], [94], [95]. Furthermore, our affordance lens reveals the central role of human-in-the-loop engagement: enterprise architects must iteratively verify, contextualize, and enrich AI-generated outputs to achieve meaningful outcomes, underscoring complementarities rather than replacements between AI and human expertise [96], [97], [98]. These insights contribute to theoretical debates on human-AI collaboration [99], [100] by specifying boundary conditions that shape when and how GenAI technologies transform EA practices, suggesting that future research should consider both micro-level user capabilities and macro-level organizational infrastructures to fully understand technology adoption and impact [101], [102], [103].

Although an overview of the affordances of GenAI in the context of EA and its use by enterprise architects to support their activities has been provided, this work is subject

to several internal **limitations**. Primarily, the research was conducted through qualitative expert interviews with employees of a single case company. Additionally, the focus was on experienced enterprise architects, who primarily perform strategic tasks, as opposed to less experienced architects, who are more involved in operative tasks. These limitations imply that not all affordances of GenAI in the context of EA may have been identified and future work could integrate additional perspectives.

The presentation of the affordances that GenAI provides to the EA and the demonstration of how enterprise architects could use GenAI for their work opens up further research opportunities. It was demonstrated that EA practices currently primarily benefit from the textual modality of GenAI. As modelling plays an important role in the context of EA, **future research** could focus on solutions enabling the generation and continual adaptation of visual models through GenAI, thereby expanding GenAI's utility beyond text. Moreover, future research could conduct comparative research across organizations, aiming to identify similarities and differences in how GenAI affordances manifest in different EA contexts. Hence, we propose a research roadmap involving multi-site and mixed-methods designs. In particular, longitudinal field experiments and multi case studies could examine quality and efficiency improvements in real-world EA tasks as GenAI integration matures. This roadmap may include both quantitative analysis (e.g., measuring performance deltas resulting from GenAI-enabled processes) and qualitative inquiries (e.g., interviews or focus groups to capture context-specific adaptations and challenges). By pursuing these directions, future research can deepen the evidence base on GenAI affordances in EA, facilitate cross-context synthesis, and inform the development of adaptable, effective practices for the evolving digital enterprise.

VI. CONCLUSION

In this paper, we demonstrated how generative AI enables support possibilities for enterprise architects. A theoretical framework was developed that presents the four affordances of *information research and synthesis*, *text generation and refinement*, *insight generation and decision support*, and *architectural content creation*. Furthermore, we demonstrate how these affordances are actualized and the specific outcomes they produce. Moreover, our findings show that GenAI (particularly through natural language prompting) functions as a machine-based cooperation partner for enterprise architects. This enables them to accelerate routine activities and gain inspiration for their work. The source material provided by GenAI serves as a basis for architects to apply their expertise to develop customized, high-quality, and innovative content in the field of EA.

APPENDIX A

CODING STRUCTURE OF DATA ANALYSIS PROCESS

See Table 4.

TABLE 4. Data structure.

Agg. Dim.	2 nd Order Theme	Exemplary 1 st Order Codes
Affordance	Information Research and Synthesis	- Identification of industry-specific business capabilities - Identification of advantages and disadvantages of specific software solutions - Identification of industry-relevant top issues (e.g., trends, competitors) - Collection of customer information
	Textual Creation and Refinement	- Content optimization of textual content (to individual preferences or target audiences) - Generation of business capability descriptions - Generation of requirements - Formulation of the architectural principles
	Insight Generation and Decision Support	- Assessment of the maturity level of a business capability - Categorization of applications according to certain criteria - Suggestions for alternative software solutions for specific business capabilities
	Architecture Content Creation	- Creation of a business capability map - Creation of a process model - Evaluation of a topic-specific PowerPoint layout - Generation of a decision matrix
Affordance Realization	Information Request Prompt	- Using a prompt to provide pros and cons of different vendors - Using a prompt to provide information about the core components of a target architecture - Using a prompt to obtain specific information from provided documents
	Evaluation Request Prompt	- Using a prompt to assess the maturity of the business capability based on specific criteria (e.g., business criticality) - Using a prompt to provide recommendations
	Optimization Request Prompt	- Using a prompt to improve the content of texts - Using a prompt to tailor the level of detail of texts to appropriate audiences - Using a prompt to request proposals for alternative software solutions
	Generation Request Prompt	- Using a prompt to generate descriptions of architectural principles - Using a prompt to transfer content from one presentation format to another - Using a prompt to generate a process model with phases, activities, and outcomes
	Tools with integrated GenAI Functions	- Generation of business capability descriptions with LeanIX - Classification of the listed application according to specific criteria within Excel - Provision of optimization suggestions based on the data stored in the database
Material Property	Natural Language Processing	- Processing and understanding textual input - Generation of coherent, specific responses based on the input
	Media Generation	- Generation of text-based content - Rephrasing of textual content

TABLE 4. (Continued.) Data structure.

	and Modification	• Generation of images
	Information Gathering	• Retrieval / provision of up-to-date information • Review and validation of the user's initial concepts
	Data Processing and Analysis	• Analysis of data for identification of potential improvement approaches • Suggestion of alternative solutions
	Structural Formatting of Content	• Creation of content in various structured formats (lists, tables) • Reformatting content from one presentation format to another

ACKNOWLEDGMENT

The authors express their sincerest gratitude to Benjamin Zeller for his expertise and guidance throughout conducting this work. The energy, time, and attention he devoted during early and late stages of this work enabled us to improve the research significantly, with many helpful suggestions and constructive feedback.

REFERENCES

- [1] N. Berente, B. Gu, J. Recker, and R. Santhanam, "Managing artificial intelligence: Special issue editor's comments," *MISQ*, vol. 45, no. 3, pp. 1433–1450, 2021.
- [2] A. Fügner, J. Grahl, A. Gupta, and W. Ketter, "Will Humans-in-the-loop become borgs? Merits and pitfalls of working with AI," *MIS Quart.*, vol. 45, no. 3, pp. 1527–1556, Sep. 2021, doi: [10.25300/misq/2021/16553](https://doi.org/10.25300/misq/2021/16553).
- [3] R. Sabherwal and V. Grover, "The societal impacts of generative artificial intelligence: A balanced perspective," *J. Assoc. Inf. Syst.*, vol. 25, no. 1, pp. 13–22, 2024, doi: [10.17705/1jais.00860](https://doi.org/10.17705/1jais.00860).
- [4] P. J. Ågerfalk, "Artificial intelligence as digital agency," *Eur. J. Inf. Syst.*, vol. 29, no. 1, pp. 1–8, Jan. 2020, doi: [10.1080/0960085x.2020.1721947](https://doi.org/10.1080/0960085x.2020.1721947).
- [5] A. Agrawal, J. S. Gans, and A. Goldfarb, "Artificial intelligence: The ambiguous labor market impact of automating prediction," *J. Econ. Perspect.*, vol. 33, no. 2, pp. 31–50, May 2019, doi: [10.1257/jep.33.2.31](https://doi.org/10.1257/jep.33.2.31).
- [6] Accenture, *What is Artificial Intelligence (AI) and Why is it Important?*. Accessed: Feb. 10, 2025. [Online]. Available: <https://www.accenture.com/us-en/insights/artificial-intelligence-summary-index>
- [7] E. Brynjolfsson, D. Li, and L. Raymond, "Generative AI at work," *Quart. J. Econ.*, vol. 140, no. 2, pp. 889–942, Apr. 2025, doi: [10.1093/qje/qjae044](https://doi.org/10.1093/qje/qjae044).
- [8] J. Hawk. (2022). *Technical Leaders Agree: AI is Now a Necessity to Compete*. Microsoft Azure Blog. Accessed: Mar. 8, 2022. [Online]. Available: <https://azure.microsoft.com/de-de/blog/technical-leaders-agree-ai-is-now-a-necessity-to-compete/>
- [9] L. Banh and G. Strobel, "Generative artificial intelligence," *Electron. Markets*, vol. 33, no. 1, pp. 1–17, Dec. 2023, doi: [10.1007/s12525-023-00680-1](https://doi.org/10.1007/s12525-023-00680-1).
- [10] G. Strobel, L. Banh, F. Möller, and T. Schoormann, "Exploring generative artificial intelligence: A taxonomy and types," in *Proc. Annu. Hawaii Int. Conf. Syst. Sci.*, Hawaii, HI, USA, 2024, pp. 1–10.
- [11] Y. K. Dwivedi et al., "Opinion paper: 'So what if ChatGPT wrote it?' multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy," *Int. J. Inf. Manage.*, vol. 71, Aug. 2023, Art. no. 102642, doi: [10.1016/j.ijinfomgt.2023.102642](https://doi.org/10.1016/j.ijinfomgt.2023.102642).
- [12] Y. A. Mohamed, A. Khanan, M. Bashir, A. H. H. M. Mohamed, M. A. E. Adiel, and M. A. Elsadiq, "The impact of artificial intelligence on language translation: A review," *IEEE Access*, vol. 12, pp. 25553–25579, 2024, doi: [10.1109/ACCESS.2024.3366802](https://doi.org/10.1109/ACCESS.2024.3366802).
- [13] Reuters. (2023). *ChatGPT Sets Record for Fastest-Growing User Base—Analyst Note*. [Online]. Available: <https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/>
- [14] A. Agrawal, J. Gans, and A. Goldfarb, *The Economics of Artificial Intelligence: An Agenda*. Chicago, IL, USA: The University of Chicago Press, 2019.
- [15] A. McAfee, D. Rock, and E. Brynjolfsson, "How to capitalize on generative AI: A guide to realizing its benefits while limiting its risks," *Harvard Bus. Rev.*, vol. 101, no. 6, pp. 43–48, 2023. [Online]. Available: <https://hbr.org/2023/11/how-to-capitalize-on-generative-ai>
- [16] T. Eloundou, S. Manning, P. Mishkin, and D. Rock, "GPTs are GPTs: Labor market impact potential of LLMs," *Science*, vol. 384, no. 6702, pp. 1306–1308, Jun. 2024, doi: [10.1126/science.adj0998](https://doi.org/10.1126/science.adj0998).
- [17] A. Chakraborty, Y. Tayob, and B. Rao. *Reinventing Enterprise Operations With Gen AI*. Accessed: Feb. 10, 2025. [Online]. Available: <https://www.accenture.com/us-en/insights/strategic-managed-services/reinvent-operations-with-genai>
- [18] A. Jadon and S. Kumar, "Leveraging generative AI models for synthetic data generation in healthcare: Balancing research and privacy," in *Proc. Int. Conf. Smart Appl., Commun. Netw. (SmartNets)*, Jul. 2023, pp. 1–4.
- [19] Statista, *Generative AI—Worldwide*. Accessed: Feb. 10, 2025. [Online]. Available: <https://www.statista.com/outlook/tmo/artificial-intelligence/generative-ai/worldwide>
- [20] Gartner, *Generative AI: What is it, Tools, Models, Applications and Use Cases*. Accessed: Feb. 10, 2025. [Online]. Available: <https://www.gartner.com/en/topics/generative-ai>
- [21] C. Ebert and P. Louridas, "Generative AI for software practitioners," *IEEE Softw.*, vol. 40, no. 4, pp. 30–38, Jul. 2023, doi: [10.1109/MS.2023.3265877](https://doi.org/10.1109/MS.2023.3265877).
- [22] L. Banh, F. Holldack, and G. Strobel, "Copiloting the future: How generative AI transforms software engineering," *Inf. Softw. Technol.*, vol. 183, Jul. 2025, Art. no. 107751, doi: [10.1016/j.infsof.2025.107751](https://doi.org/10.1016/j.infsof.2025.107751).
- [23] T. Härlin, G. B. Rova, A. Singla, O. Sokolov, and A. Sukharevsky. (2023). *Exploring Opportunities in the Generative AI Value Chain*. McKinsey Company. [Online]. Available: <https://www.mckinsey.com/capabilities/quantumblack/our-insights/exploring-opportunities-in-the-generative-ai-value-chain>
- [24] B. Meskó and E. J. Topol, "The imperative for regulatory oversight of large language models (or generative AI) in healthcare," *Npj Digit. Med.*, vol. 6, no. 1, p. 120, Jul. 2023, doi: [10.1038/s41746-023-00873-0](https://doi.org/10.1038/s41746-023-00873-0).
- [25] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," *Nature Med.*, vol. 29, no. 8, pp. 1930–1940, Aug. 2023, doi: [10.1038/s41591-023-02448-8](https://doi.org/10.1038/s41591-023-02448-8).
- [26] N. Sheppard. *AI-Powered Enterprise Architecture: Defining The Future*. Accessed: Feb. 10, 2025. [Online]. Available: <https://www.leanix.net/en/blog/ai-powered-enterprise-architecture>
- [27] F. Ahlemann, E. Stettiner, M. Messerschmidt, and C. Legner, *Strategic Enterprise Architecture Management: Challenges, Best Practices and Future Developments*, 1st ed., Berlin, Germany: Springer, 2012.
- [28] T. Tamm, P. B. Seddon, G. Shanks, and P. Reynolds, "How does enterprise architecture add value to organisations?" *Commun. Assoc. Inf. Syst.*, vol. 28, no. 1, pp. 141–168, 2011, doi: [10.17705/1cais.02810](https://doi.org/10.17705/1cais.02810).
- [29] G. Vial, "Understanding digital transformation: A review and a research agenda," *J. Strategic Inf. Syst.*, vol. 28, no. 2, pp. 118–144, Jun. 2019, doi: [10.1016/j.jsis.2019.01.003](https://doi.org/10.1016/j.jsis.2019.01.003).
- [30] S. Noy and W. Zhang, "Experimental evidence on the productivity effects of generative artificial intelligence," *Science*, vol. 381, no. 6654, pp. 187–192, Jul. 2023, doi: [10.1126/science.adh2586](https://doi.org/10.1126/science.adh2586).
- [31] F. Dell'Acqua, E. McFowland, E. R. Mollick, H. Lifshitz-Assaf, K. Kellogg, S. Rajendran, L. Kraye, F. Candelon, and K. R. Lakhani, "Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality," *Harvard Bus. School, Work. Paper* 24-013, Sep. 2023. [Online]. Available: <https://www.hbs.edu/faculty/Pages/item.aspx?num=64700>
- [32] J. D. Rittelmeyer and K. Sandkuhl, "Effects of artificial intelligence on enterprise architectures—A structured literature review," in *Proc. IEEE 25th Int. Enterprise Distrib. Object Comput. Workshop (EDOCW)*, Oct. 2021, pp. 130–137.
- [33] L. Fitriani, M. L. Khodra, and K. Surendro, "TOGAF-based enterprise architecture framework for utilizing artificial intelligence," in *Proc. Int. Conf. Comput., Control, Informat. Appl. (IC3INA)*, Bandung, Indonesia, Oct. 2023, pp. 90–95.
- [34] D. Denni-Fiberesima, "Navigating the generative ai-enabled enterprise architecture landscape: Critical success factors for AI adoption and strategic integration," in *Networks and Systems, Navigating the Technological Tide: The Evolution and Challenges of Business Model Innovation*. Cham, Switzerland: Springer, 2024, pp. 210–222.

- [35] O. Volkoff and D. M. Strong, "Critical realism and affordances: Theorizing IT-associated organizational change processes," *MIS Quart.*, vol. 37, no. 3, pp. 819–834, Sep. 2013.
- [36] M. Hakimi, M. S. Zarinkhail, and S. Z. Musawi, "Exploring the fusion of enterprise architecture, blockchain, and AI in digital governance: A systematic review," *Int. J. Softw. Eng. Comput. Sci.*, vol. 4, no. 2, pp. 497–511, Aug. 2024, doi: [10.35870/ijsecs.v4i2.2832](https://doi.org/10.35870/ijsecs.v4i2.2832).
- [37] M. D. Myers and M. Newman, "The qualitative interview in IS research: Examining the craft," *Inf. Org.*, vol. 17, no. 1, pp. 2–26, Jan. 2007, doi: [10.1016/j.infoandorg.2006.11.001](https://doi.org/10.1016/j.infoandorg.2006.11.001).
- [38] P. M. Leonardi, "When flexible routines meet flexible technologies: Affordance, constraint, and the imbrication of human and material agencies," *MIS Quart.*, vol. 35, no. 1, p. 147, 2011, doi: [10.2307/23043493](https://doi.org/10.2307/23043493).
- [39] T. B. Brown, "Language models are few-shot learners," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, vol. 33, 2022, pp. 1877–1901.
- [40] A. Oussidi and A. Elhassouny, "Deep generative models: Survey," in *Proc. Int. Conf. Intell. Syst. Comput. Vis.*, Apr. 2018, pp. 1–8.
- [41] G. Yenduri, M. Ramalingam, G. C. Selvi, Y. Supriya, G. Srivastava, P. K. R. Maddikunta, G. D. Raj, R. H. Jhaveri, B. Prabadevi, W. Wang, A. V. Vasilakos, and T. R. Gadekallu, "GPT (generative pre-trained transformer)—A comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions," *IEEE Access*, vol. 12, pp. 54608–54649, 2024, doi: [10.1109/ACCESS.2024.3389497](https://doi.org/10.1109/ACCESS.2024.3389497).
- [42] M. Lankhorst, *Enterprise Architecture at Work*. Berlin, Germany: Springer, 2009.
- [43] F. Gampfer, A. Jürgens, M. Müller, and R. Buchkremer, "Past, current and future trends in enterprise architecture—A view beyond the horizon," *Comput. Ind.*, vol. 100, pp. 70–84, Sep. 2018, doi: [10.1016/j.compind.2018.03.006](https://doi.org/10.1016/j.compind.2018.03.006).
- [44] H. Jonkers, M. M. Lankhorst, H. W. L. ter Doest, F. Arbab, H. Bosma, and R. J. Wieringa, "Enterprise architecture: Management tool and blueprint for the organisation," *Inf. Syst. Frontiers*, vol. 8, no. 2, pp. 63–66, Feb. 2006, doi: [10.1007/S10796-006-7970-2](https://doi.org/10.1007/S10796-006-7970-2).
- [45] E. Niemi, "Enterprise architecture benefits: Perceptions from literature and practice," in *Proc. 7th Int. Bus. Inf. Manage. Assoc. (IBIMA) Conf. Internet Inf. Syst. Digit. Age*, K. S. Soliman, Ed., Norristown, PA, USA: IBIMA Publishing, 2006.
- [46] S. Feuerriegel, J. Hartmann, C. Janiesch, and P. Zschech, "Generative AI," *Bus. Inf. Syst. Eng.*, vol. 66, no. 1, pp. 111–126, Feb. 2024, doi: [10.1007/s12599-023-00834-7](https://doi.org/10.1007/s12599-023-00834-7).
- [47] S. Goyal, F. Hussein, I. Thukral, and T. Trimble, "Empowering enterprise architecture: Leveraging NLP for time efficiency and strategic alignment," in *Proc. Syst. Inf. Eng. Design Symp. (SIEDS)*, Charlottesville, VA, USA, May 2024, pp. 431–436.
- [48] H. Guo and S. Gao, "AI application in enterprise architecture management: From the perspective of tools," in *Proc. ICEB, Zhuhai, China*, 2024, pp. 615–621.
- [49] L. Görgen, E. Müller, M. Triller, B. Nast, and K. Sandkuhl, "Large language models in enterprise modeling: Case study and experiences," in *Proc. 12th Int. Conf. Model-Based Softw. Syst. Eng.*, Rome, Italy, 2024, pp. 74–85.
- [50] D. Chen, G. Doumeings, and F. Vernadat, "Architectures for enterprise integration and interoperability: Past, present and future," *Comput. Ind.*, vol. 59, no. 7, pp. 647–659, Sep. 2008, doi: [10.1016/j.compind.2007.12.016](https://doi.org/10.1016/j.compind.2007.12.016).
- [51] K. Sandkuhl, B. S. Barn, and S. Barat, "Neural text generators in enterprise modeling: Can chatgpt be used as proxy domain expert?" in *Proc. Int. Conf. Inf. Syst. Develop.*, Lisbon, Portugal, Oct. 2023, pp. 1–12.
- [52] C. Ziche and G. Apruzzese, "LLM4PM: A case study on using large language models for process modeling in enterprise organizations," in *Business Information Processing, Business Process Management: Blockchain, Robotic Process Automation, Central and Eastern European, Educators and Industry Forum*. Cham, Switzerland: Springer, 2024, pp. 472–483.
- [53] B. S. Barn, S. Barat, and K. Sandkuhl, "Adaptation of enterprise modeling methods for large language models," in *Business Information Processing, The Practice of Enterprise Modeling*. Cham, Switzerland: Springer, sec. 2024, pp. 3–18.
- [54] P.-A. Kolev, H. H. Pruss, J. R. Wilken, and K. Sandkuhl, "Grass-root enterprise modelling: How large language models can help," in *Business Information Processing, The Practice of Enterprise Modeling*. Cham, Switzerland: Springer, 2024, pp. 123–139.
- [55] C. Engel, P. Ebel, and J. M. Leimeister, "Cognitive automation," *Electron. Markets*, vol. 32, no. 1, pp. 339–350, Mar. 2022, doi: [10.1007/s12525-021-00519-7](https://doi.org/10.1007/s12525-021-00519-7).
- [56] D. Simon, K. Fischbach, and D. Schoder, "An exploration of enterprise architecture research," *Commun. Assoc. Inf. Syst.*, vol. 32, no. 1, pp. 1–72, 2013, doi: [10.17705/icaais.03201](https://doi.org/10.17705/icaais.03201).
- [57] D. Goerzig and T. Bauernhansl, "Enterprise architectures for the digital transformation in small and medium-sized enterprises," *Proc. CIRP*, vol. 67, pp. 540–545, Sep. 2018, doi: [10.1016/j.procir.2017.12.257](https://doi.org/10.1016/j.procir.2017.12.257).
- [58] N. E. Kamalabai, I. Donoghue, and L. Hannola, "Sustainable enterprise architecture: A critical imperative for substantiating artificial intelligence," in *Proc. Portland Int. Conf. Manage. Eng. Technol. (PICMET)*, Portland, OR, USA, Aug. 2024, pp. 1–12.
- [59] J. Yang, H. Jin, R. Tang, X. Han, Q. Feng, H. Jiang, S. Zhong, B. Yin, and X. Hu, "Harnessing the power of LLMs in practice: A survey on ChatGPT and beyond," *ACM Trans. Knowl. Discovery Data*, vol. 18, no. 6, pp. 1–32, Jul. 2024, doi: [10.1145/3649506](https://doi.org/10.1145/3649506).
- [60] H. K. Chaubey, G. Tripathi, R. Ranjan, and S. K. Gopalaiyengar, "Comparative analysis of RAG, fine-tuning, and prompt engineering in chatbot development," in *Proc. Int. Conf. Future Technol. Smart Soc. (ICFTSS)*, Kuala Lumpur, Malaysia, Aug. 2024, pp. 169–172.
- [61] P. Stecher, M. Pohl, and K. Turowski, "Enterprise architecture's effects on organizations' ability to adopt artificial intelligence—A resource-based perspective," in *Proc. ECIS*, 2020, pp. 1–16.
- [62] D. Russo, "Navigating the complexity of generative AI adoption in software engineering," *ACM Trans. Softw. Eng. Methodol.*, vol. 33, no. 5, pp. 1–50, Jun. 2024, doi: [10.1145/3652154](https://doi.org/10.1145/3652154).
- [63] M. Pinski and A. Benlian, "AI literacy for users—A comprehensive review and future research directions of learning methods, components, and effects," *Comput. Hum. Behav., Artif. Hum.*, vol. 2, no. 1, Jan. 2024, Art. no. 100062, doi: [10.1016/j.chbah.2024.100062](https://doi.org/10.1016/j.chbah.2024.100062).
- [64] M. A. K. Raiaan, M. S. H. Mukta, K. Fatema, N. M. Fahad, S. Sakib, M. M. J. Mim, J. Ahmad, M. E. Ali, and S. Azam, "A review on large language models: Architectures, applications, taxonomies, open issues and challenges," *IEEE Access*, vol. 12, pp. 26839–26874, 2024, doi: [10.1109/ACCESS.2024.3365742](https://doi.org/10.1109/ACCESS.2024.3365742).
- [65] C. Dremel, M. M. Herterich, J. Wulf, and J. vom Brocke, "Actualizing big data analytics affordances: A revelatory case study," *Inf. Manage.*, vol. 57, no. 1, Jan. 2020, Art. no. 103121, doi: [10.1016/j.im.2018.10.007](https://doi.org/10.1016/j.im.2018.10.007).
- [66] O. Volkoff and D. M. Strong, "Affordance theory and how to use it in IS research," in *Routledge Companions in Business, Management and Accounting, The Routledge Companion to Management Information Systems*, R. Galliers and M.-K. Stein, Eds., Evanston, IL, USA: Routledge, 2018, pp. 232–245.
- [67] D. M. Strong, O. Volkoff, S. A. Johnson, L. R. Pelletier, B. Tulu, I. Bar-On, J. Trudel, and L. Garber, "A theory of organization-EHR affordance actualization," *J. Assoc. Inf. Syst.*, vol. 15, no. 2, p. 2, 2014.
- [68] D. E. Leidner, E. Gonzalez, and H. Koch, "An affordance perspective of enterprise social media and organizational socialization," *J. Strategic Inf. Syst.*, vol. 27, no. 2, pp. 117–138, Jun. 2018, doi: [10.1016/j.jsis.2018.03.003](https://doi.org/10.1016/j.jsis.2018.03.003).
- [69] D. Shin, "The actualization of meta affordances: Conceptualizing affordance actualization in the metaverse games," *Comput. Hum. Behav.*, vol. 133, Aug. 2022, Art. no. 107292, doi: [10.1016/j.chb.2022.107292](https://doi.org/10.1016/j.chb.2022.107292).
- [70] C. Lehrer, A. Wieneke, J. vom Brocke, R. Jung, and S. Seidel, "How big data analytics enables service innovation: Materiality, affordance, and the individualization of service," *J. Manage. Inf. Syst.*, vol. 35, no. 2, pp. 424–460, Apr. 2018, doi: [10.1080/07421222.2018.1451953](https://doi.org/10.1080/07421222.2018.1451953).
- [71] W. Du, S. L. Pan, D. E. Leidner, and W. Ying, "Affordances, experimentation and actualization of FinTech: A blockchain implementation study," *J. Strategic Inf. Syst.*, vol. 28, no. 1, pp. 50–65, Mar. 2019, doi: [10.1016/j.jsis.2018.10.002](https://doi.org/10.1016/j.jsis.2018.10.002).
- [72] R. Keller, A. Stohr, G. Fridgen, J. Lockl, and A. Rieger, "Affordance-experimentation-actualization theory in artificial intelligence research—A predictive maintenance story," in *Proc. ICIS*, vol. 1, 2019, pp. 1–17.
- [73] U. Schultze and M. Avital, "Designing interviews to generate rich data for information systems research," *Inf. Org.*, vol. 21, no. 1, pp. 1–16, Jan. 2011, doi: [10.1016/j.infoandorg.2010.11.001](https://doi.org/10.1016/j.infoandorg.2010.11.001).
- [74] H. Kallio, A.-M. Pietilä, M. Johnson, and M. Kangasniemi, "Systematic methodological review: Developing a framework for a qualitative semi-structured interview guide," *J. Adv. Nursing*, vol. 72, no. 12, pp. 2954–2965, Dec. 2016, doi: [10.1111/jan.13031](https://doi.org/10.1111/jan.13031).

- [75] H. J. Rubin and I. Rubin, *Qualitative Interviewing: The Art of Hearing Data*, 2nd ed., Newbury Park, CA, USA: Sage, 2005.
- [76] R. K. Yin, *Case Study Research and Applications: Design and Methods*, 6th ed., Thousand Oaks, CA, USA: SAGE, 2018.
- [77] K. M. Eisenhardt, "Building theories from case study research," *Acad. Manage. Rev.*, vol. 14, no. 4, p. 532, Oct. 1989, doi: [10.2307/258557](https://doi.org/10.2307/258557).
- [78] A. L. Strauss and J. M. Corbin, *Grounded Theory in Practice*. Newbury Park, CA, USA: Sage, 1997.
- [79] D. A. Gioia, K. G. Corley, and A. L. Hamilton, "Seeking qualitative rigor in inductive research," *Organizational Res. Methods*, vol. 16, no. 1, pp. 15–31, Jan. 2013, doi: [10.1177/1094428112452151](https://doi.org/10.1177/1094428112452151).
- [80] C. Urquhart, H. Lehmann, and M. D. Myers, "Putting the 'theory' back into grounded theory: Guidelines for grounded theory studies in information systems," *Inf. Syst. J.*, vol. 20, no. 4, pp. 357–381, Jul. 2010, doi: [10.1111/j.1365-2575.2009.00328.x](https://doi.org/10.1111/j.1365-2575.2009.00328.x).
- [81] B. G. Glaser and A. L. Strauss, *Grounded Theory: Strategien Qualitativer Forschung*, 3rd ed., Bern, Switzerland: Verlag Hans Huber, 2010.
- [82] P. Angelov, E. Soares, R. Jiang, N. I. Arnold, and P. M. Atkinson, "Explainable artificial intelligence: An analytical review," *WIREs Data Min. Knowl.*, vol. 11, no. 5, pp. 1–13, 2021, doi: [10.1002/widm.1424](https://doi.org/10.1002/widm.1424).
- [83] P. West, X. Lu, N. Dziri, F. Brahman, L. Li, J. D. Hwang, L. Jiang, J. Fisher, A. Ravichander, K. Chandu, B. Newman, P. W. Koh, A. Ettinger, and Y. Choi, "The generative AI paradox: What it can create, it may not understand," 2024, *arXiv:2311.00059*.
- [84] M. O. Riedl, "Human-centered artificial intelligence and machine learning," *Hum. Behav. Emerg. Technol.*, vol. 1, no. 1, pp. 33–36, Jan. 2019, doi: [10.1002/hbe2.117](https://doi.org/10.1002/hbe2.117).
- [85] Y. Wu, "Integrating generative AI in education: How ChatGPT brings challenges for future learning and teaching," *J. Adv. Res. Educ.*, vol. 2, no. 4, pp. 6–10, Sep. 2023, doi: [10.56397/jare.2023.07.02](https://doi.org/10.56397/jare.2023.07.02).
- [86] F. F.-H. Nah, R. Zheng, J. Cai, K. Siau, and L. Chen, "Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration," *J. Inf. Technol. Case Appl. Res.*, vol. 25, no. 3, pp. 277–304, Jul. 2023, doi: [10.1080/15228053.2023.2233814](https://doi.org/10.1080/15228053.2023.2233814).
- [87] I. Pesovski, R. Santos, R. Henriques, and V. Trajkovic, "Generative AI for customizable learning experiences," *Sustainability*, vol. 16, no. 7, p. 3034, Apr. 2024, doi: [10.3390/su16073034](https://doi.org/10.3390/su16073034).
- [88] Y. Huang, J. Xu, J. Lai, Z. Jiang, T. Chen, Z. Li, Y. Yao, X. Ma, L. Yang, H. Chen, S. Li, and P. Zhao, "Advancing transformer architecture in long-context large language models: A comprehensive survey," 2023, *arXiv:2311.12351*.
- [89] J. Kaddour, J. Harris, M. Mozes, H. Bradley, R. Raileanu, and R. McHardy, "Challenges and applications of large language models," 2023, *arXiv:2307.10169*.
- [90] N. Mungoli, "Scalable, distributed AI frameworks: Leveraging cloud computing for enhanced deep learning performance and efficiency," 2023, *arXiv:2304.13738*.
- [91] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, W. Ye, Y. Zhang, Y. Chang, P. S. Yu, Q. Yang, and X. Xie, "A survey on evaluation of large language models," *ACM Trans. Intell. Syst. Technol.*, vol. 15, no. 3, pp. 1–45, Jun. 2024, doi: [10.1145/3641289](https://doi.org/10.1145/3641289).
- [92] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, Dec. 2023, doi: [10.1145/3571730](https://doi.org/10.1145/3571730).
- [93] P. Mikalef, K. Conboy, J. E. Lundström, and A. Popovič, "Thinking responsibly about responsible AI and 'the dark side' of AI," *Eur. J. Inf. Syst.*, vol. 31, no. 3, pp. 257–268, May 2022, doi: [10.1080/0960085x.2022.2026621](https://doi.org/10.1080/0960085x.2022.2026621).
- [94] A. Baiyere, H. Salmela, H. Nieminen, and T. Kankainen, "Assessing digital capabilities for digital transformation—The MIND framework," *Inf. Syst. J.*, vol. 35, no. 1, pp. 6–38, Jan. 2025, doi: [10.1111/isj.12519](https://doi.org/10.1111/isj.12519).
- [95] O. Ali, P. A. Murray, S. Muhammed, Y. K. Dwivedi, and S. Rashiti, "Evaluating organizational level IT innovation adoption factors among global firms," *J. Innov. Knowl.*, vol. 7, no. 3, Jul. 2022, Art. no. 100213, doi: [10.1016/j.jik.2022.100213](https://doi.org/10.1016/j.jik.2022.100213).
- [96] W. Wang, G. Gao, and R. Agarwal, "Friend or foe? Teaming between artificial intelligence and workers with variation in experience," *Manag. Sci.*, vol. 70, no. 9, pp. 5753–5775, 2023, doi: [10.1287/mnsc.2021.00588](https://doi.org/10.1287/mnsc.2021.00588).
- [97] C. Anthony, B. A. Bechky, and A.-L. Fayard, "'Collaborating' with AI: Taking a system view to explore the future of work," *Org. Sci.*, vol. 34, no. 5, pp. 1672–1694, Sep. 2023, doi: [10.1287/orsc.2022.1651](https://doi.org/10.1287/orsc.2022.1651).
- [98] T. Grønsund and M. Aanestad, "Augmenting the algorithm: Emerging human-in-the-loop work configurations," *J. Strategic Inf. Syst.*, vol. 29, no. 2, Jun. 2020, Art. no. 101614, doi: [10.1016/j.jsis.2020.101614](https://doi.org/10.1016/j.jsis.2020.101614).
- [99] S. Raisch and S. Krakowski, "Artificial intelligence and management: The automation–augmentation paradox," *Acad. Manage. Rev.*, vol. 46, no. 1, pp. 192–210, Jan. 2021, doi: [10.5465/amr.2018.0072](https://doi.org/10.5465/amr.2018.0072).
- [100] S. Raisch and K. Fomina, "Combining human and artificial intelligence: Hybrid problem-solving in organizations," *Acad. Manage. Rev.*, vol. 50, no. 2, pp. 441–464, Apr. 2025, doi: [10.5465/amr.2021.0421](https://doi.org/10.5465/amr.2021.0421).
- [101] M.-K. Stein and A. Shollo, "Microfoundations of rationality in the age of AI: On emotions, bodies and intelligence," *Inf. Org.*, vol. 35, no. 3, Sep. 2025, Art. no. 100583, doi: [10.1016/j.infoandorg.2025.100583](https://doi.org/10.1016/j.infoandorg.2025.100583).
- [102] L. Hillebrand, S. Raisch, and J. Schad, "Managing with artificial intelligence: An integrative framework," *Acad. Manage. Ann.*, vol. 19, no. 1, pp. 343–375, Jan. 2025, doi: [10.5465/annals.2022.0072](https://doi.org/10.5465/annals.2022.0072).
- [103] K. Haki, D. Safaei, A. Magan, and M. Griffiths, "Integrating generative AI into enterprise platforms: Insights from salesforce," *Inf. Syst. J.*, vol. 35, no. 5, pp. 1497–1512, Sep. 2025, doi: [10.1111/isj.12593](https://doi.org/10.1111/isj.12593).



LEONARDO BANH is currently a Research Associate and a Ph.D. Student with the Chair of Information Systems and Software Engineering, University of Duisburg–Essen, Germany. His research focuses on generative AI, particularly its sociotechnical implications for individuals and organizations, human-GenAI collaboration, and the evolution from GenAI toward agentic information systems.



TUAN KHANG TRAN is currently a Strategy Consultant with a focus on an Enterprise Architecture at an internationally operating IT consulting firm. He draws on his academic background in information systems and provides strategic guidance and consulting services to organizations on their IT transformation journeys, helping them align technology and business strategy for sustainable success.



GERO STROBEL is currently an Assistant Professor with the University of Duisburg–Essen and the Deputy Professor of the Chair of Information Systems, University of Stuttgart, Germany. His research has been published in several journals and major conference proceedings in the field of information systems. His research focuses on the design of information systems and ecosystems, particularly in the field of human-centered generative artificial intelligence and human-GenAI collaboration.

...